console.log( )

alert( )

APNA COLLEGE

String Methods

Methods - actions that can be performed on objects.

Format

stringName.method( )

JS

APNA COLLEGE

String Methods

let msg = "   Hello   ";

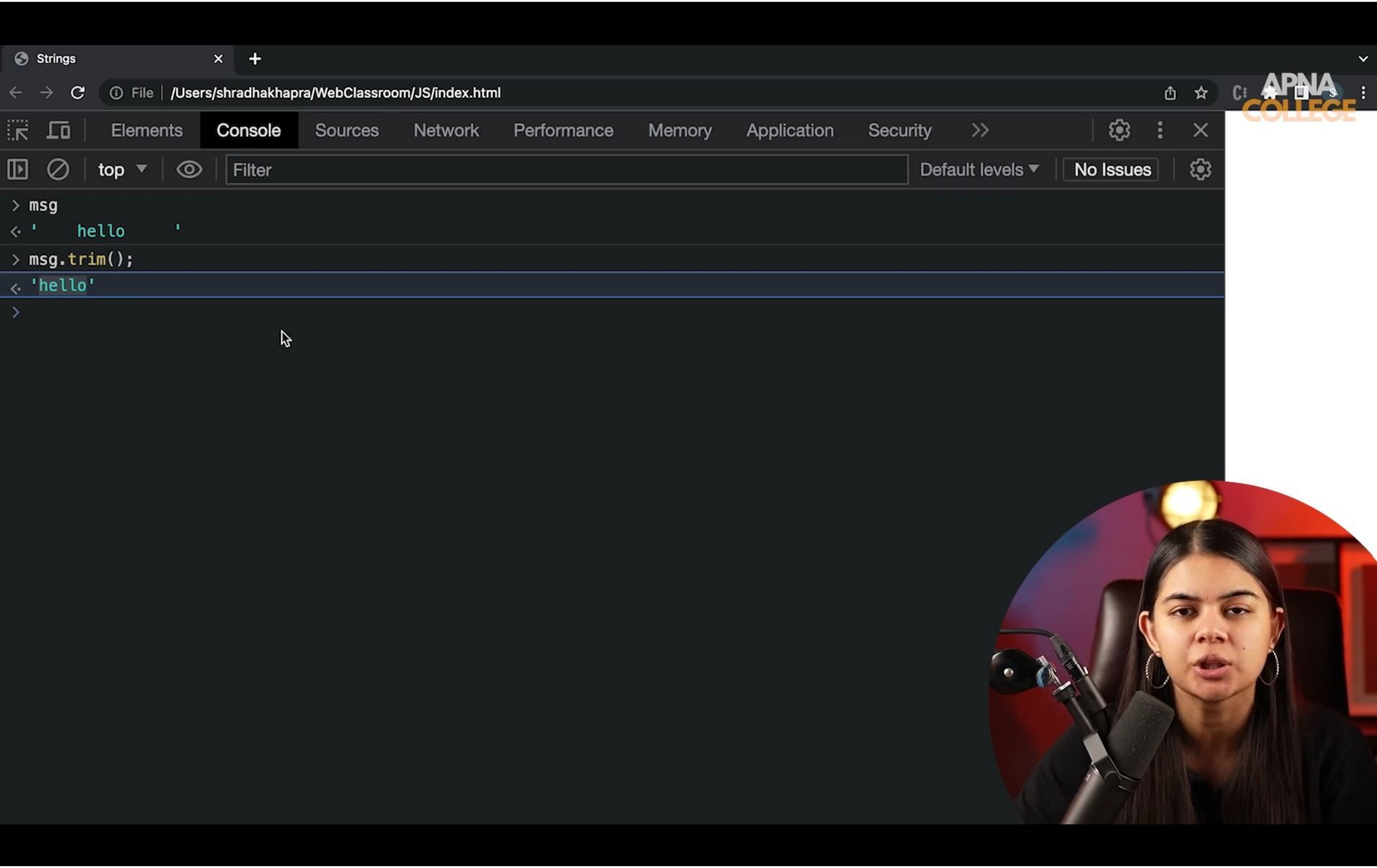
str.trim( )

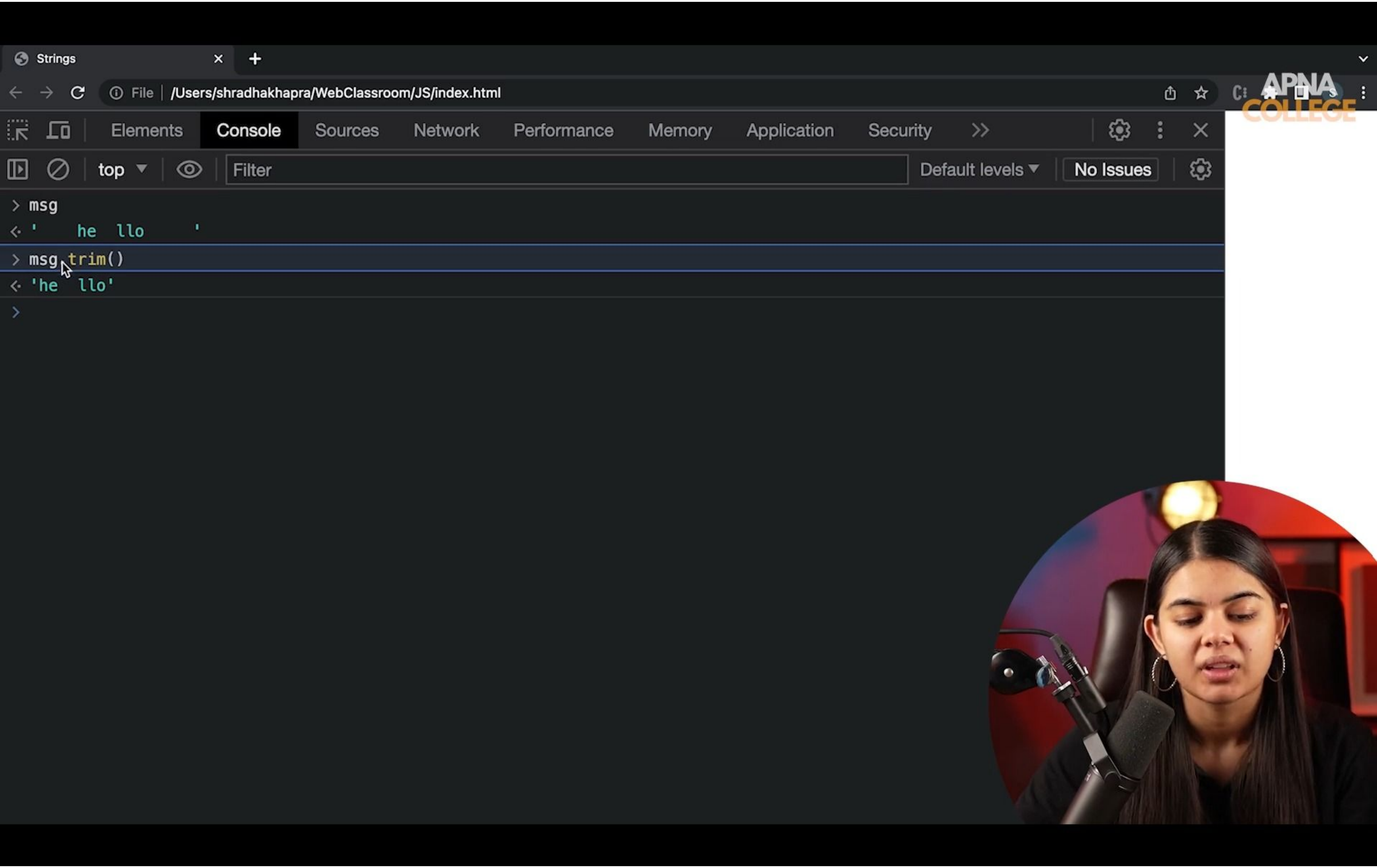
Trims whitespaces from both ends of string & returns a new one.

```
> let msg = "    Hello    ";
<· undefined
> msg.trim();
<· 'Hello'
> msg
<· '    Hello    '
```

output : "Hello", but value of msg remains same.

JS


Strings
File
/Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
> msg
<· '   hello   '
> msg.trim();
<· 'hello'


Strings
/Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
> msg
<· '   he llo   '
> msg.trim()
<· 'he llo'

index.html | app.js

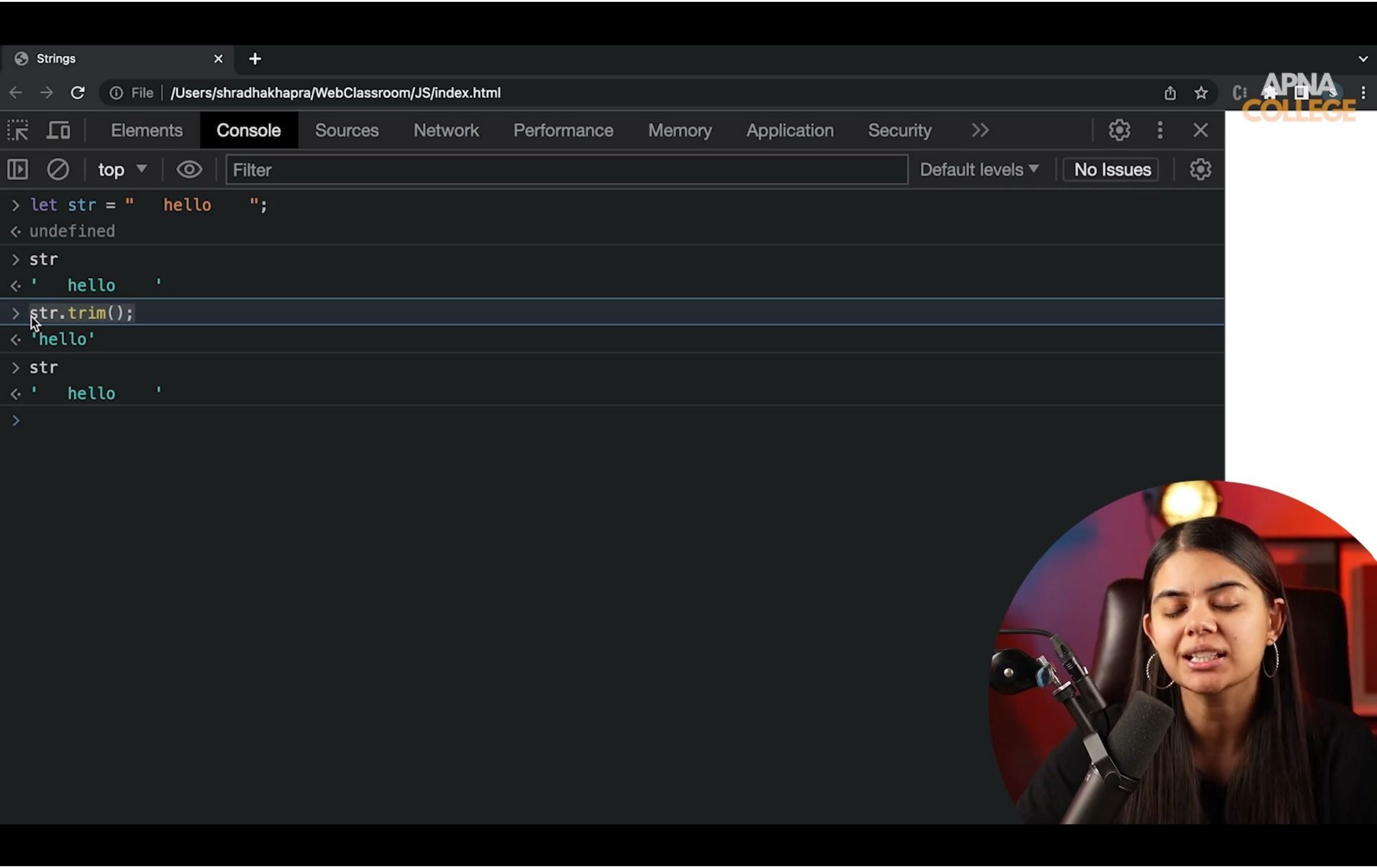
app.js > newPass

```javascript
// let msg = "    he  llo     ";

let password = prompt("set your password");
let newPass = password.trim();
console.log(password);

```

APNA COLLEGE


Strings
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
let str = "   hello   ";
undefined
str
'   hello   '
str.trim();
'hello'
str
'   hello   '

String are Immutable in JS

No changes can be made to strings.

Whenever we do try to make a change, a new string is created and old one remains same.

String Methods

```
let str = "Random string";
```

```
str.toUpperCase( )        "RANDOM STRING"

str.toLowerCase( )        "random string"
```

Strings

/Users/shradhakhapra/WebClassroom/JS/index.html

Elements | Console | Sources | Network | Performance | Memory | Application | Security

top | Filter | Default levels | No Issues

```
> name
< 'Apna College'
> name.toLowerCase();
< 'apna college'
> name.toLowerCase
< f toLowerCase() { [native code] }
> name.toUpperCase();
< 'APNA COLLEGE'
>
```

APNA COLLEGE

String Methods with Arguments

Argument is a some value that we pass to the method.

Format

stringName.method(arg)

let name = "shradha"

h

JS

indexOf

Returns the **first index of occurrence** of some value in string. Or gives -1 if not found.

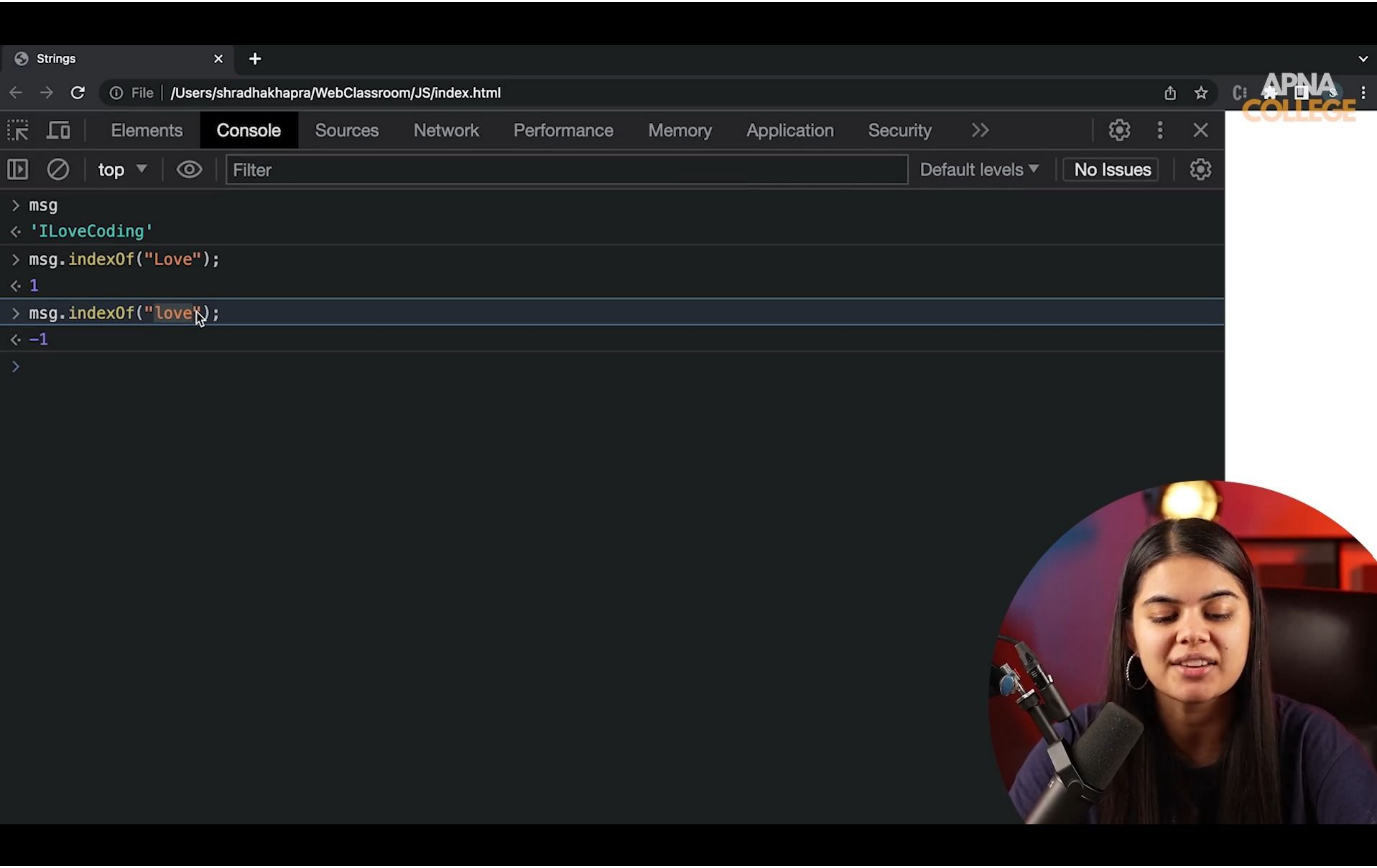
```
let str = "IloveCoding";

str.indexOf("love")      // 1

str.indexOf("J")         // -1 (not found)

str.indexOf("o")         // 2 (only 1 index)
```

JS


Strings
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
> msg
< 'ILoveCoding'
> msg.indexOf("Love");
< 1
> msg.indexOf("love");
< -1

Method Chaining

Using one method after another. Order of execution will be left to right.

```
str.toUpperCase( ).trim( )
```

index.html app.js

app.js > ...

```javascript
let msg = "    hello   ";
// let newMsg = msg.trim();
// console.log("after trim : ", newMsg);
// newMsg = newMsg.toUpperCase();
// console.log("after uppercase : ", newMsg);
let newMsg = msg.trim().toUpperCase();
```

APNA COLLEGE

02:08 / 02:48

slice

Returns a part of the original string as a new string.

```
let str = "IloveCoding";

    str.slice(5)              // "Coding"

    str.slice(1, 4)           // "love"

    str.slice(-num)   =   str.slice(length-num)
```

JS

str.slice(start, end)

1, 5

[1-4]

APNA COLLEGE

slice

Returns a part of the original string as a new string.

```
let str = "IloveCoding";
```

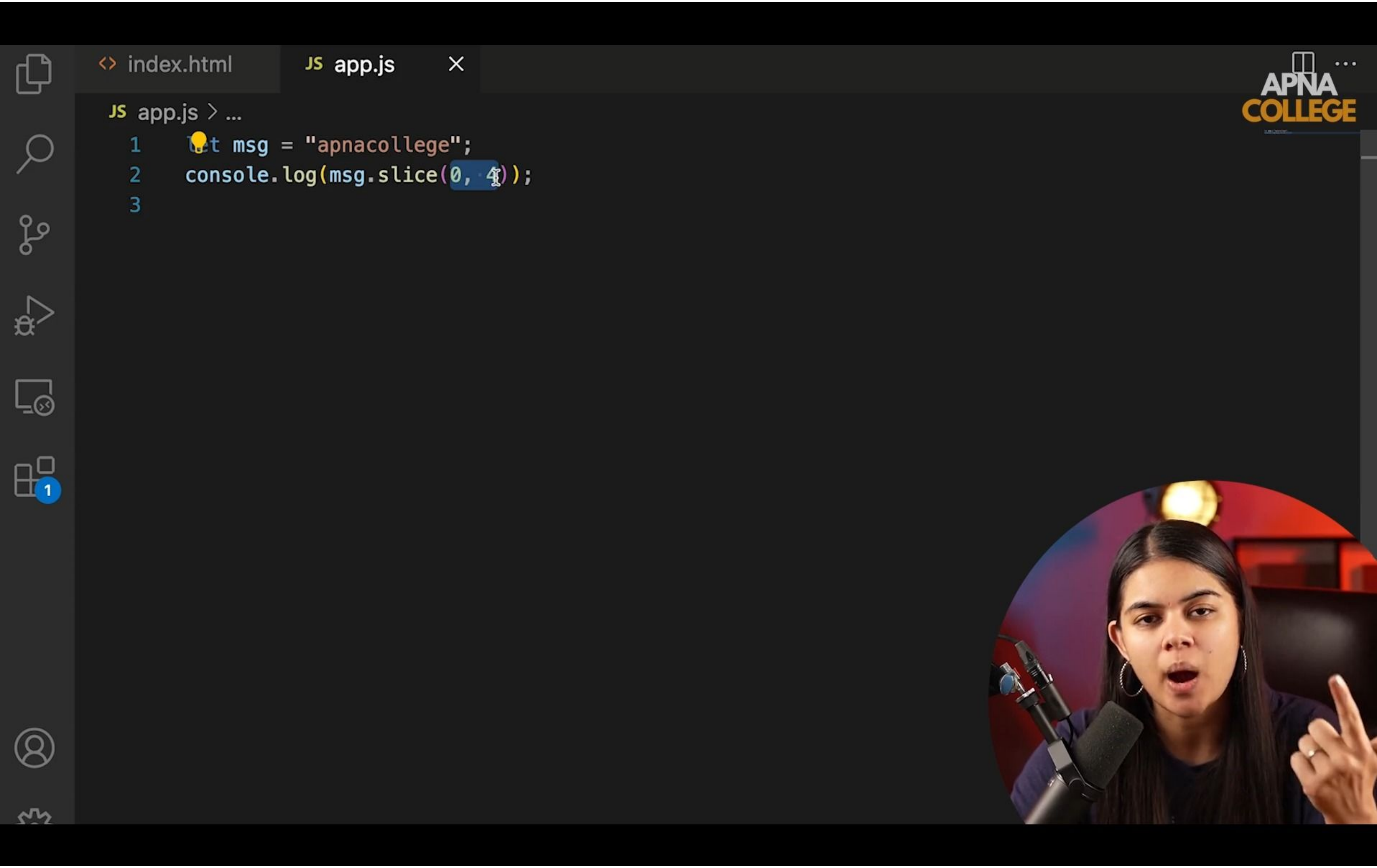
0 1

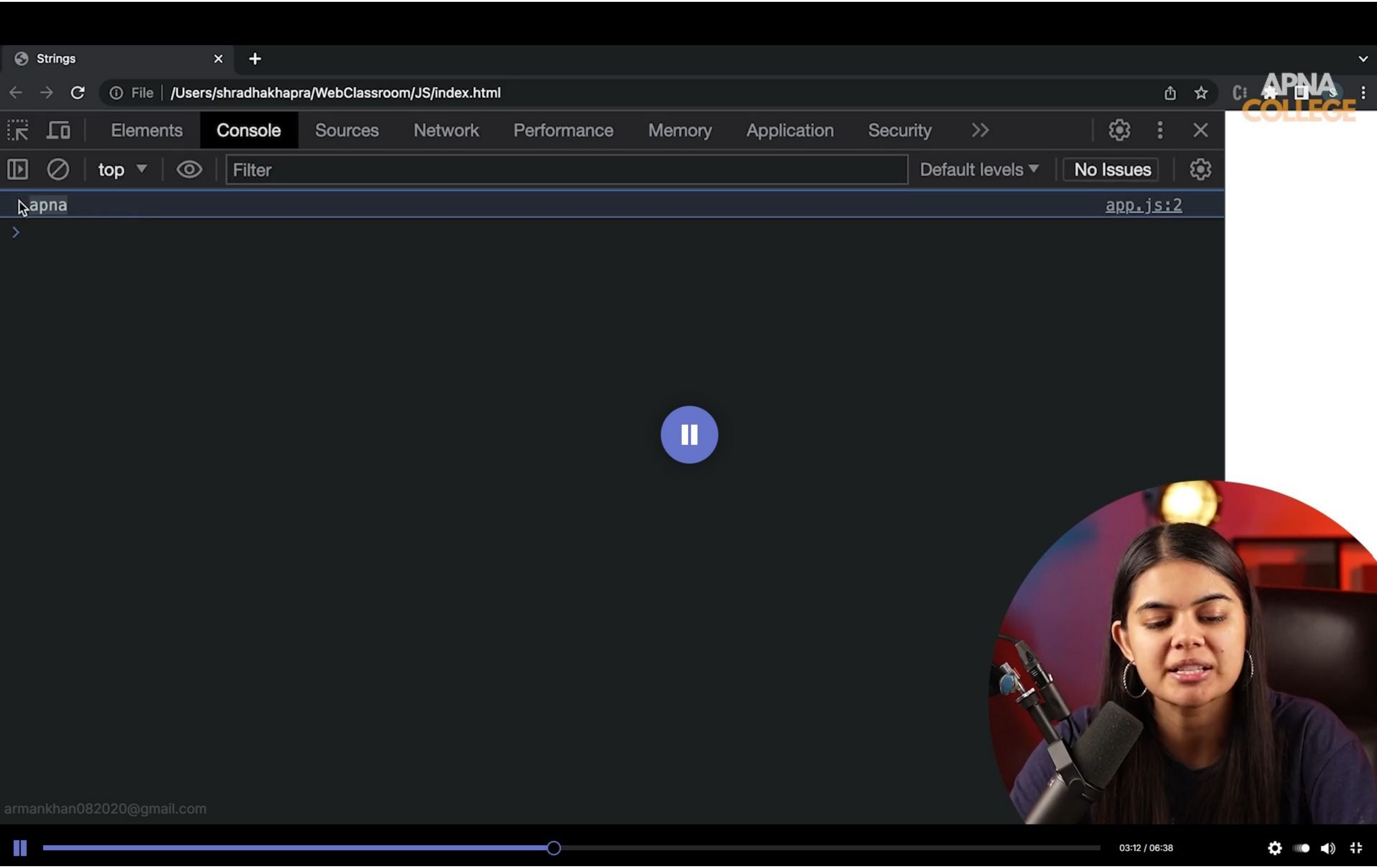
```
str.slice(5)              // "Coding"

str.slice(1, 4)           // "love"

str.slice(-num)   =   str.slice(length-num)
```

JS


index.html
JS app.js
JS app.js > ...
let msg = "apnacollege";
console.log(msg.slice(0, 4));
APNA COLLEGE
1


Strings
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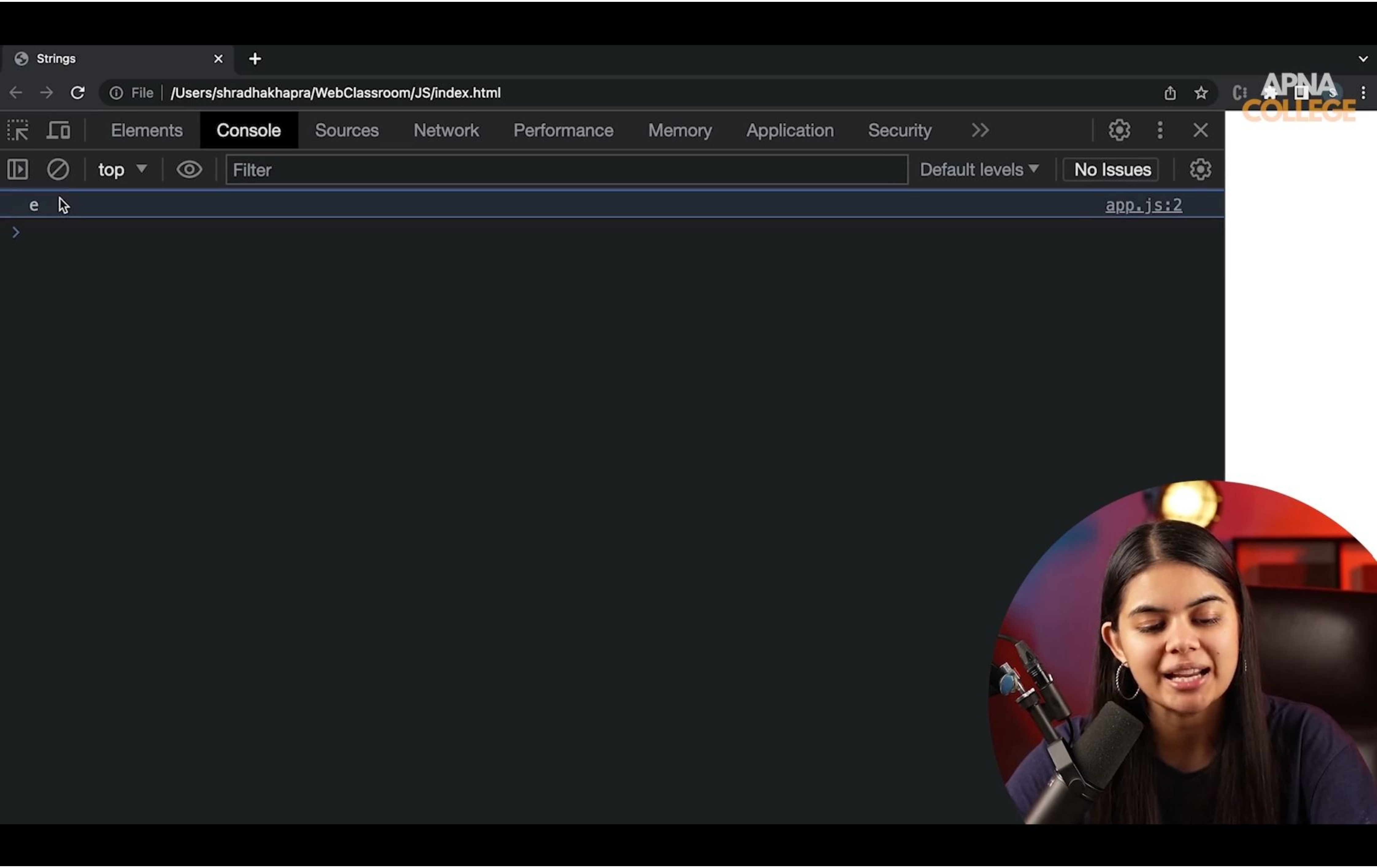
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
apna
app.js:2
armankhan082020@gmail.com
03:12 / 06:38

index.html   app.js

app.js > ...

```js
let msg = "apnacollege";
console.log(msg.slice(-1)); //11-1 => 10
```


Strings
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
e
app.js:2

slice

Returns a part of the original string as a new string.

```
let str = "IloveCoding";

str.slice(5)            // "Coding"

str.slice(1, 4)         // "love"

str.slice(-num)  =  str.slice(length-num)
```

str.slice (start, end)

1, 5.

str.slice (start)

end = str.length

str.slice (-ve)

-2

[str.length - 2]

JS

APNA COLLEGE

replace

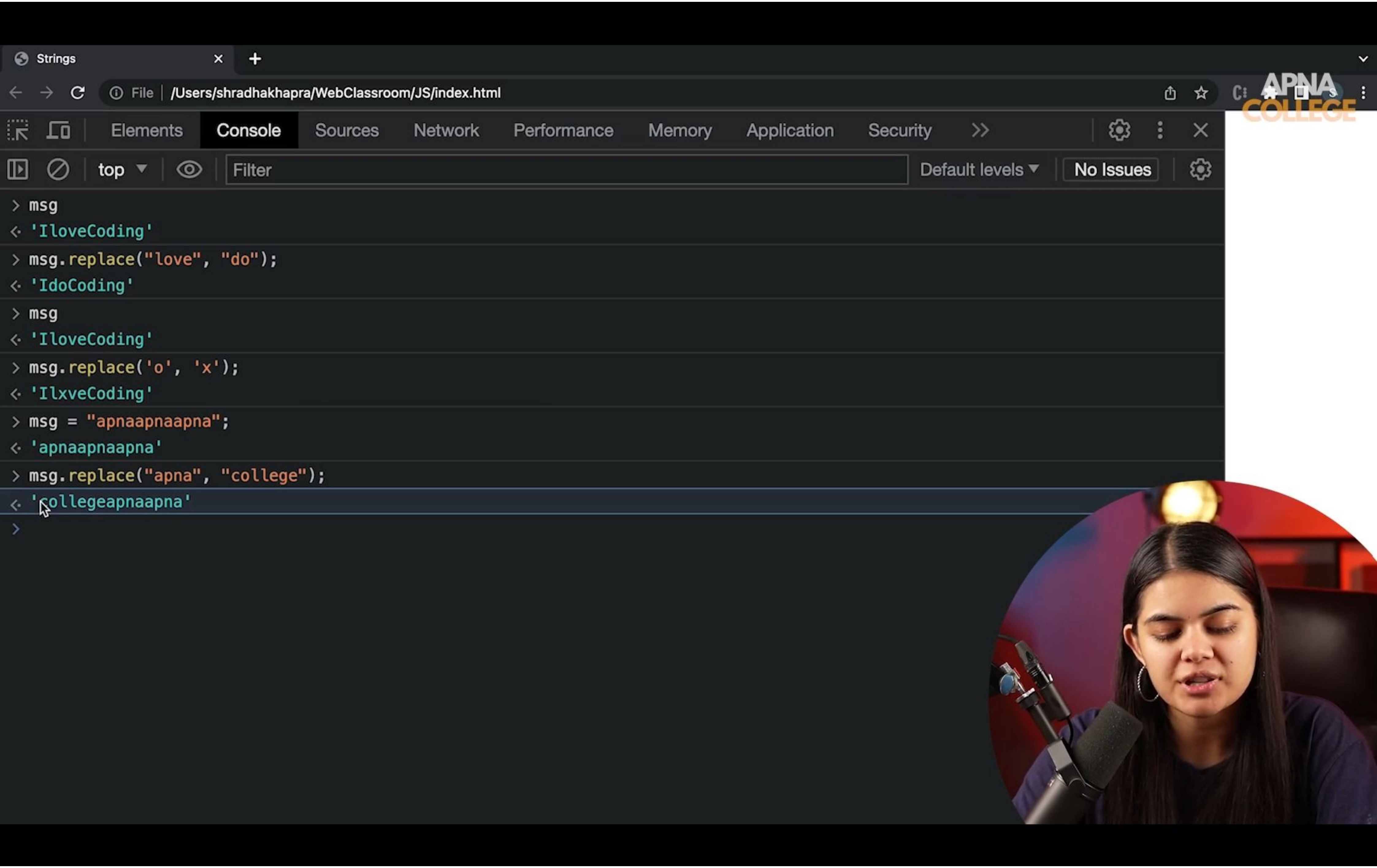
Searches a value in the string & returns a new string with the value replaced.

```
let str = "IloveCoding";

    str.slice("love", "do")    // "IdoCoding"
              val      new

    str.slice("o", "x")        // "IlxveCoding"
```

JS


Strings
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
> msg
< 'IloveCoding'
> msg.replace("love", "do");
< 'IdoCoding'
> msg
< 'IloveCoding'
> msg.replace('o', 'x');
< 'IlxveCoding'
> msg = "apnaapnaapna";
< 'apnaapnaapna'
> msg.replace("apna", "college");
< 'collegeapnaapna'

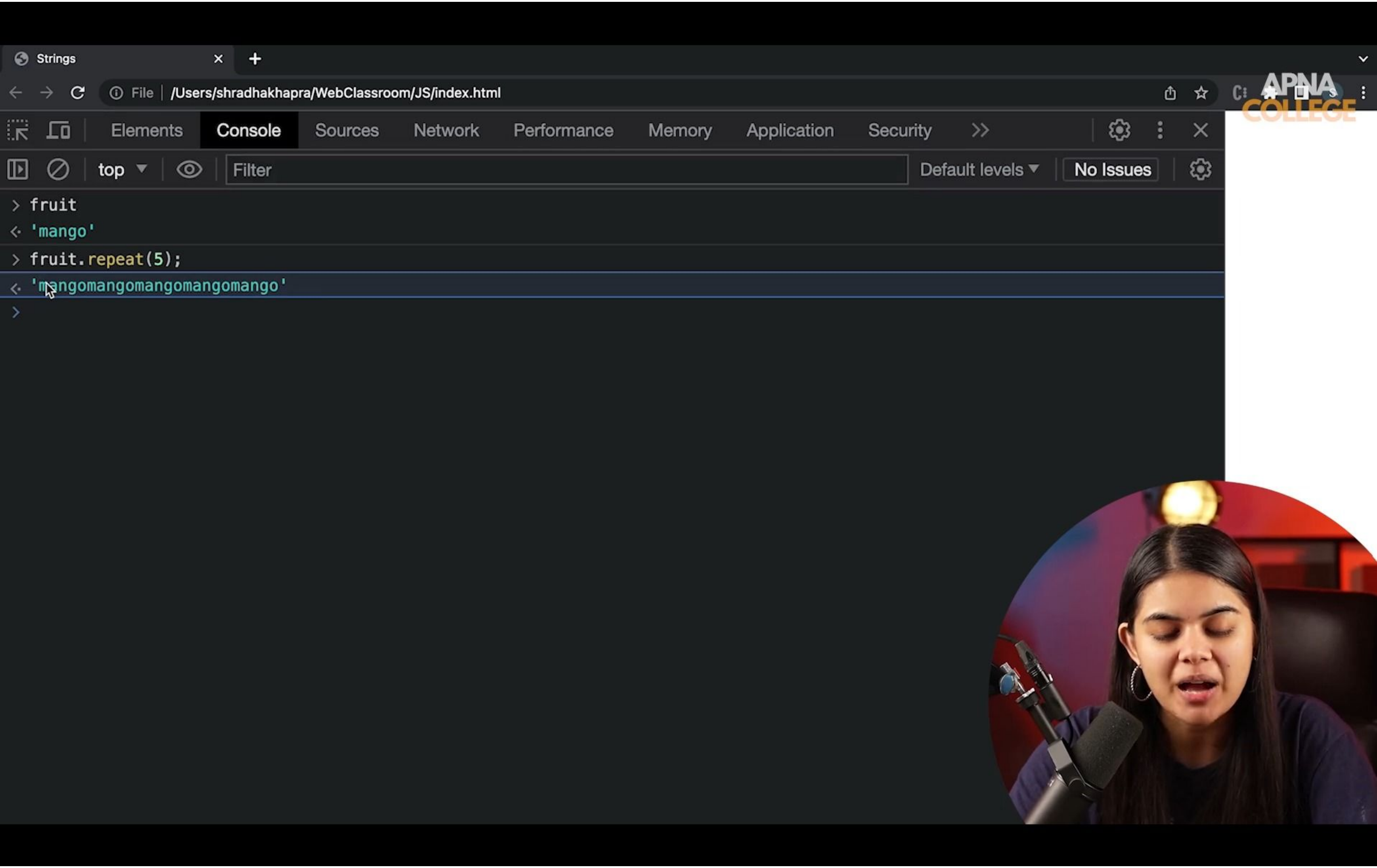
repeat

Returns a string with the number of copies of a string

```
let str = "Mango";

    str.repeat(3)          // "MangoMangoMango"
```

JS


Strings
/Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
No Issues
APNA COLLEGE
> fruit
< 'mango'
> fruit.repeat(5);
< 'mangomangomangomangomango'

Practice Qs

Qs. For the Give String :

```
let msg = "help!";
```

Trim it & convert it to uppercase.

Qs. For the String -> let name = "ApnaCollege", predict the output for following :

name.slice(4, 9)

name.indexOf("na")

name.replace("Apna","Our")

Qs. Separate the "College" part in above string & replace 'l' with 't' in it.

APNA COLLEGE

Practice Qs

msg.trim().toUpperCase();

Qs. For the Give String :

```
let msg = "help!";
```

Trim it & convert it to uppercase.

Qs. For the String -> let name = "ApnaCollege", predict the output for following :

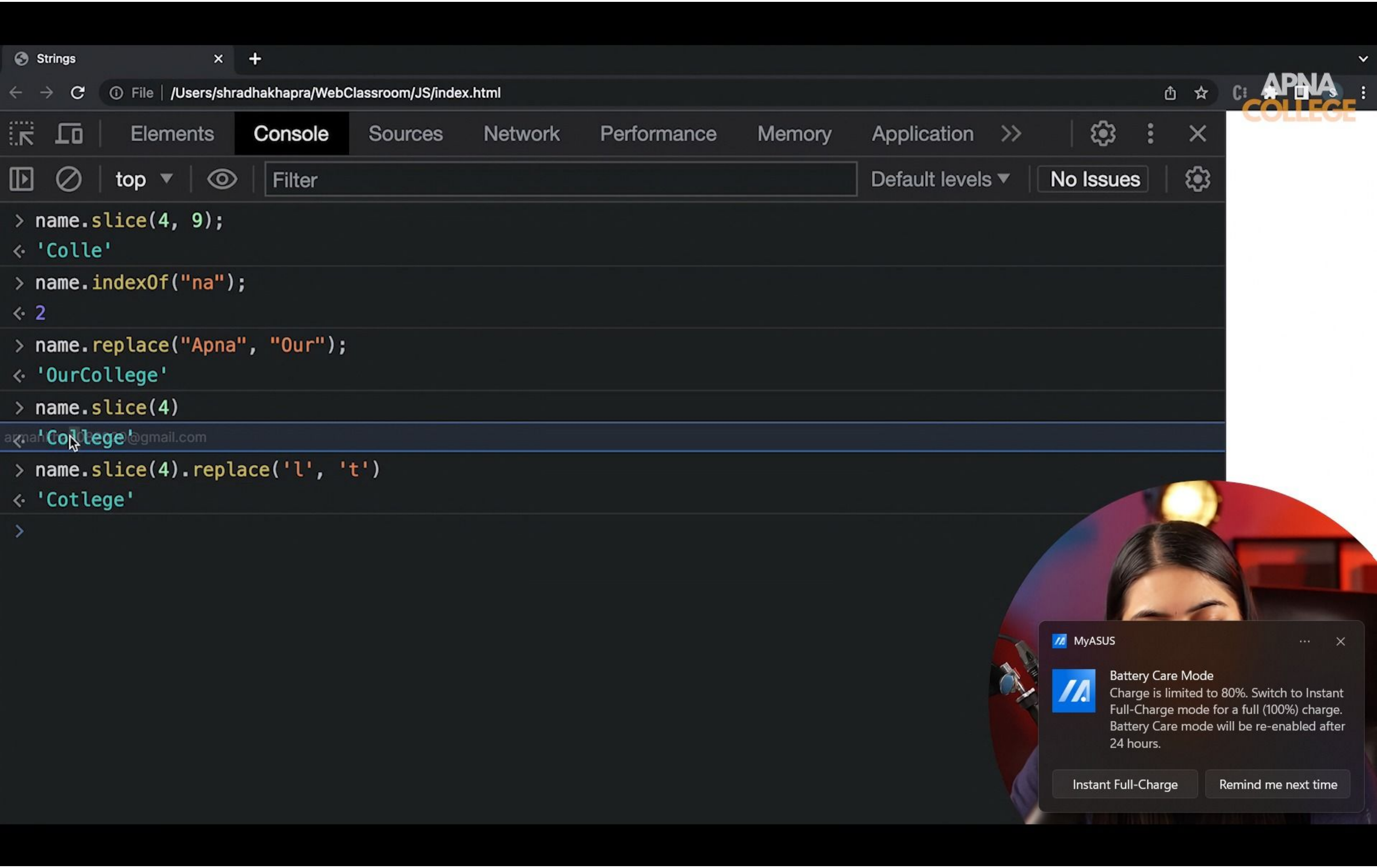
0 1 2 3 4 5 6 7 8 9 10

name.slice(4, 9)   "Colle"

name.indexOf("na")   2

name.replace("Apna","Our")

Qs. Separate the "College" part in above string & replace 'l' with 't' in it.



Strings
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Memory
Application
top
Filter
Default levels
No Issues
name.slice(4, 9);
'Colle'
name.indexOf("na");
2
name.replace("Apna", "Our");
'OurCollege'
name.slice(4)
'College'
name.slice(4).replace('l', 't')
'Cotlege'
MyASUS
Battery Care Mode
Charge is limited to 80%. Switch to Instant Full-Charge mode for a full (100%) charge. Battery Care mode will be re-enabled after 24 hours.
Instant Full-Charge
Remind me next time

Array (Data Structure)

Linear collection of things

JS

Array (Data Structure)

Linear collection of things

30

50

```
let student1 = "aman";
let student2 = "shradha";
let student3 = "rajat";
```

student1[0]

```
let students = ["aman", "shradha", "rajat"]
```

JS

JS

File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console Sources Network Performance

top Filter Default levels No Issues

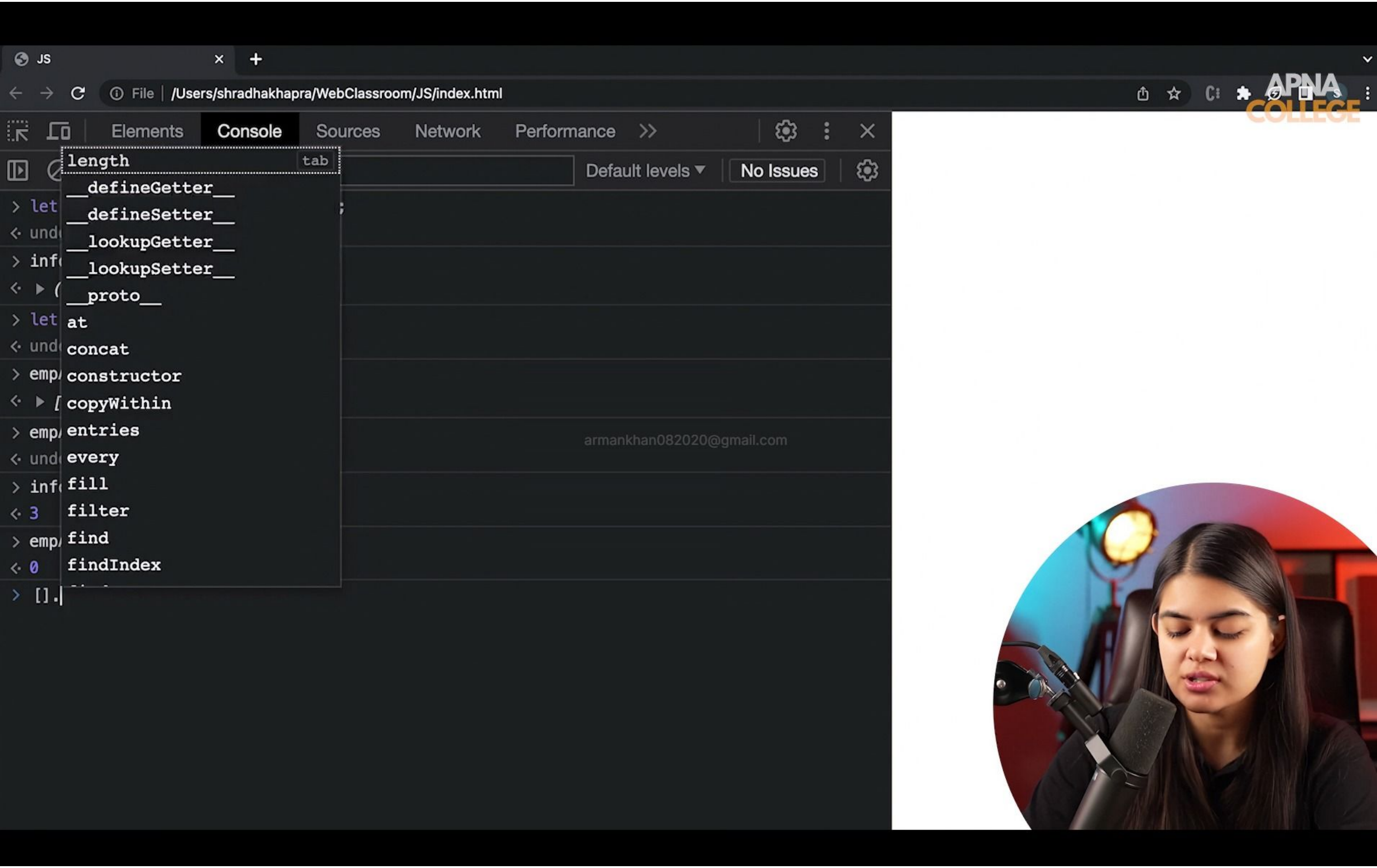
```
> let nums = [2, 4, 6, 8];
< undefined
> nums
< ▶ (4) [2, 4, 6, 8]
> let name = "shradha";
< undefined
> name[0]
< 's'
> nums[0]
< 2
> nums[1]
< 4
> nums[2]
< 6
> nums[3]
< 8
> nums[4]
< undefined
> nums.length
< 4
> typeof nums
< 'object'
>
```

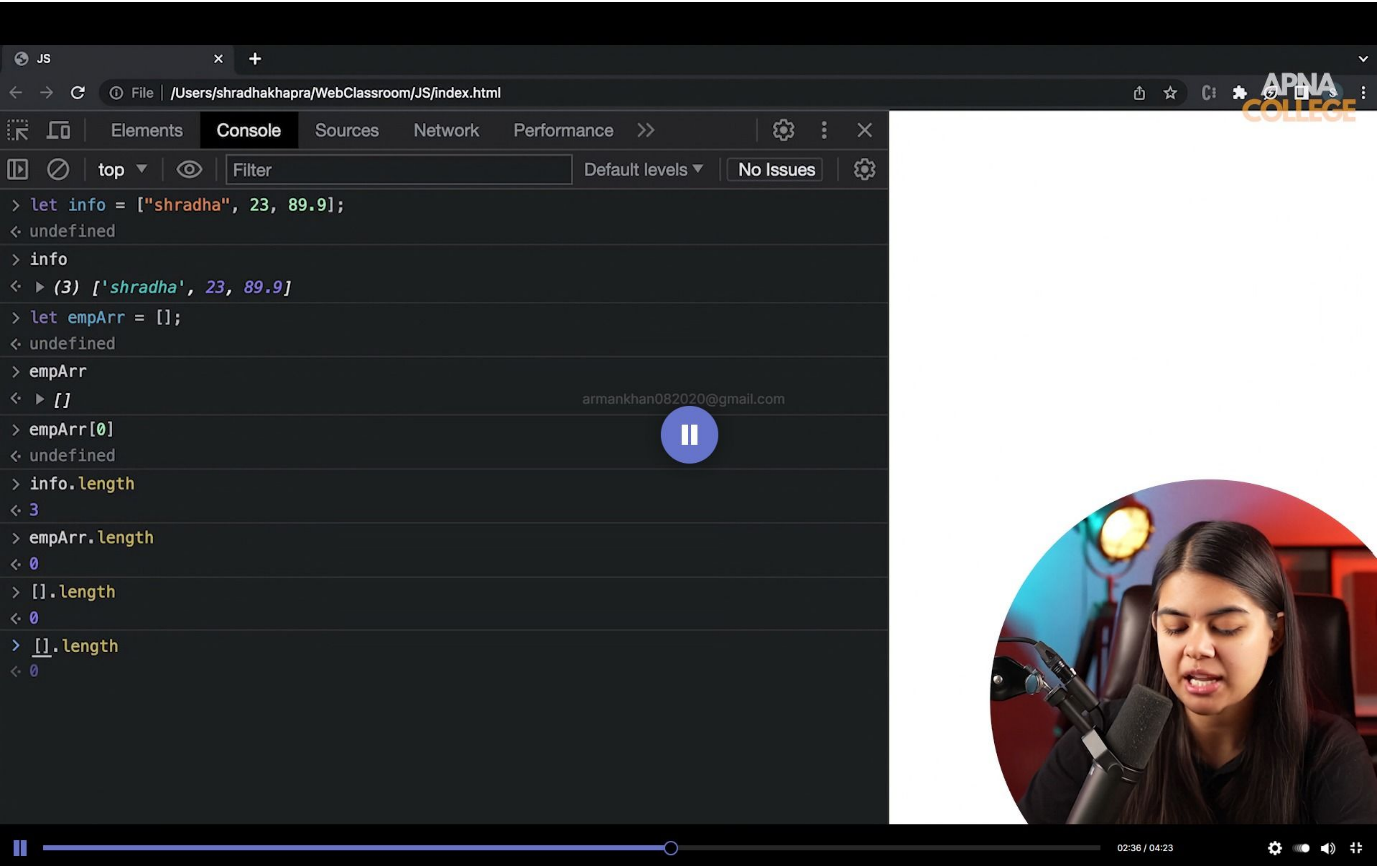
Creating Arrays

```
let marks = [99, 85, 93, 76, 62];
let names = ["adam", "bob", "catlyn"];
let info = ["aman", 25, 6.1];   //mixed array

//empty array
let newArr = [];
```

arr.length


JS
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
Default levels
No Issues
length tab
__defineGetter__
__defineSetter__
__lookupGetter__
__lookupSetter__
__proto__
at
concat
constructor
copyWithin
entries
every
fill
filter
find
findIndex
let
infe
let
emp
emp
infe
3
emp
0
[].
armankhan082020@gmail.com


JS
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
> let info = ["shradha", 23, 89.9];
< undefined
> info
< (3) ['shradha', 23, 89.9]
> let empArr = [];
< undefined
> empArr
< []
> empArr[0]
< undefined
> info.length
< 3
> empArr.length
< 0
> [].length
< 0
> [].length
< 0
armankhan082020@gmail.com
02:36 / 04:23

```
> let info = ["shradha", 23, 89.9];
<· undefined
> info
<· ▶ (3) ['shradha', 23, 89.9]
> let empArr = [];
<· undefined
> empArr
<· ▶ []
> empArr[0]
<· undefined
> info.length
<· 3
> empArr.length
<· 0
> [].length
<· 0
> [1, 2, 3, 4].length
<· 4
>
```


JS

File /Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console Sources Network Performance

top Filter Default levels No Issues

```
> let info = ["shradha", 23, 89.9];
< undefined
> info
< ▶ (3) ['shradha', 23, 89.9]
> let empArr = [];
< undefined
> empArr
< ▶ []
> empArr[0]
< undefined
> info.length
< 3
> empArr.length
< 0
> [].length
< 0
> [1, 2, 3, 4].length
< 4
> info[0]
< 'shradha'
> info[0][0]
< 's'
```


immutable

Arrays are Mutable

```
> let fruits = ["mango", "apple", "litchi"];
<· undefined
> fruits[0] = "banana";
<· 'banana'
> fruits
<· ▶ (3) ['banana', 'apple', 'litchi']
```

JS

File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console Sources Network Performance >>

top Filter Default levels No Issues

```
> let name = "rohit";
<· undefined
> name[0] = 'm';
<· 'm'
> name
<· 'rohit'
> let fruits = ["mango", "apple", "litchi"];
<· undefined
> fruits
<· ▶ (3) ['mango', 'apple', 'litchi']
> fruits[0] = "banana";
<· 'banana'
> fruits
<· ▶ (3) ['banana', 'apple', 'litchi']
>
```

COLLEGE

Elements Console Sources Network Performance

top Filter Default levels No Issues

```
> let name = "rohit";
< undefined
> name[0] = 'm';
< 'm'
> name
< 'rohit'
> let fruits = ["mango", "apple", "litchi"];
< undefined
> fruits
< ▶ (3) ['mango', 'apple', 'litchi']
> fruits[0] = "banana";
< 'banana'
> fruits
< ▶ (3) ['banana', 'apple', 'litchi']
> fruits[1] = "pineapple";
< 'pineapple'
> fruits
< ▶ (3) ['banana', 'pineapple', 'litchi']
> fruits[10] = "papaya";
< 'papaya'
> fruits
< ▶ (11) ['banana', 'pineapple', 'litchi', empty × 7, 'papaya']
>
```

APNA COLLEGE

Array Methods

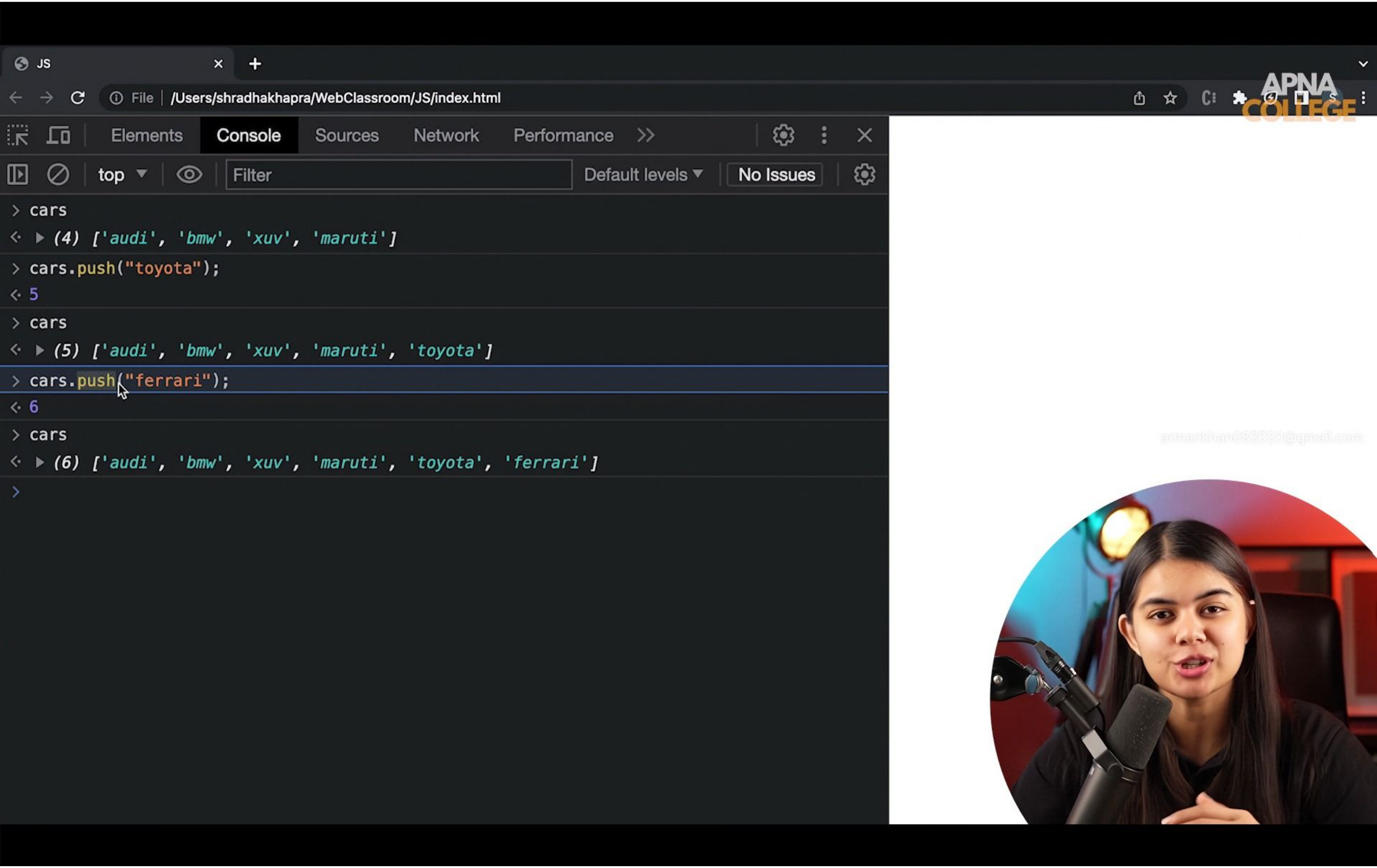
Push : add to end

Pop : delete from end & returns it

Unshift : add to start

Shift : delete from start & returns it

JS

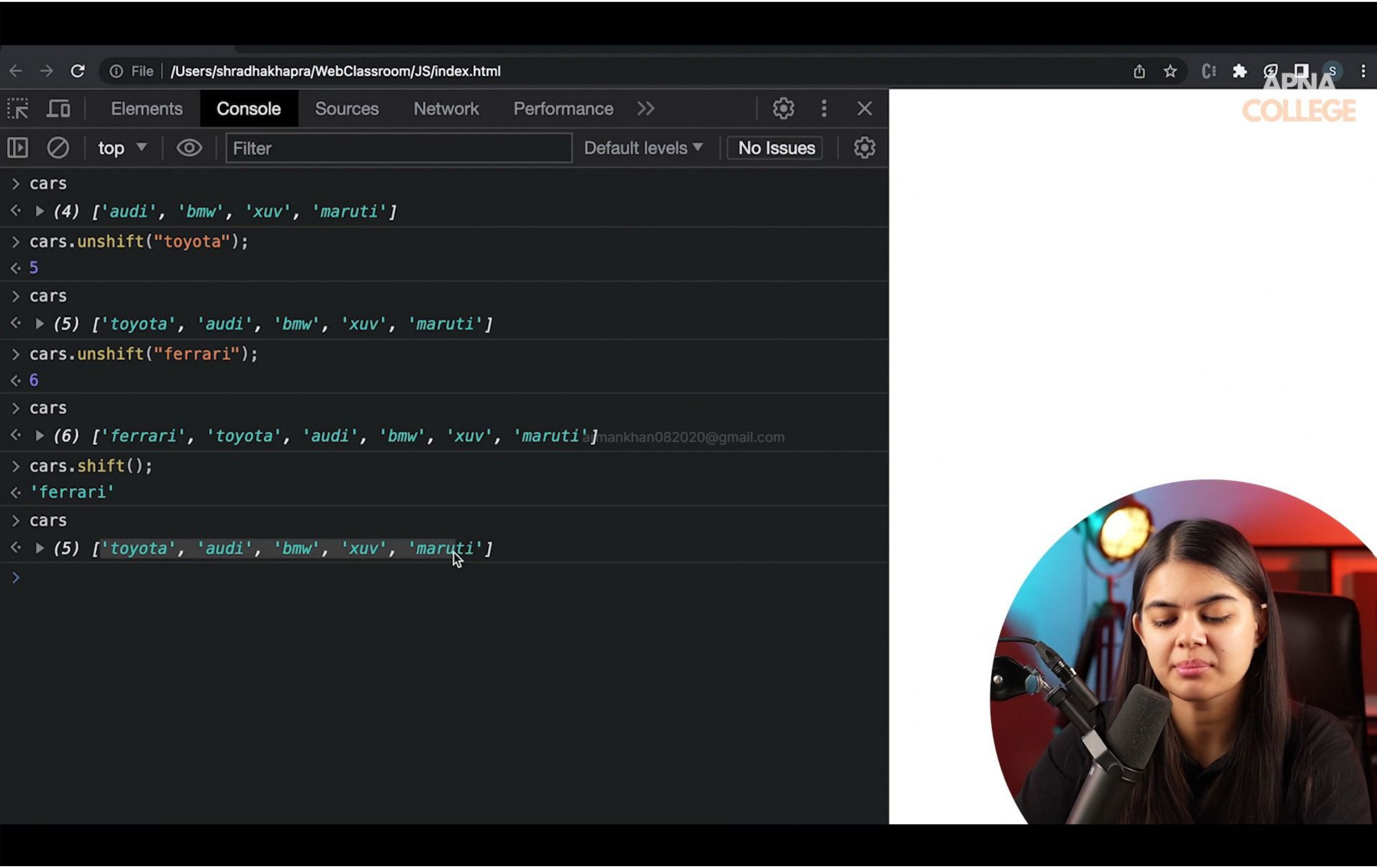

JS
File | /Users/shradhakhapra/WebClassroom/JS/index.html
APNA COLLEGE
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
> cars
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.push("toyota");
< 5
> cars
< ▶ (5) ['audi', 'bmw', 'xuv', 'maruti', 'toyota']
> cars.push("ferrari");
< 6
> cars
< ▶ (6) ['audi', 'bmw', 'xuv', 'maruti', 'toyota', 'ferrari']
>

File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console Sources Network Performance

top Filter Default levels No Issues

```
> cars
< ▸ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.push("toyota");
< 5
> cars
< ▸ (5) ['audi', 'bmw', 'xuv', 'maruti', 'toyota']
> cars.push("ferrari");
< 6
> cars
< ▸ (6) ['audi', 'bmw', 'xuv', 'maruti', 'toyota', 'ferrari']
> cars.pop();
< 'ferrari'
> cars
< ▸ (5) ['audi', 'bmw', 'xuv', 'maruti', 'toyota']
> cars.pop();
< 'toyota'
> cars
< ▾ (4) ['audi', 'bmw', 'xuv', 'maruti']
    0: "audi"
    1: "bmw"
    2: "xuv"
    3: "maruti"
    length: 4
  ▸ [[Prototype]]: Array(0)
```



/Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
> cars
< (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.unshift("toyota");
< 5
> cars
< (5) ['toyota', 'audi', 'bmw', 'xuv', 'maruti']
> cars.unshift("ferrari");
< 6
> cars
< (6) ['ferrari', 'toyota', 'audi', 'bmw', 'xuv', 'maruti']
> cars.shift();
< 'ferrari'
> cars
< (5) ['toyota', 'audi', 'bmw', 'xuv', 'maruti']
APNA
COLLEGE

File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> cars
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.unshift("toyota");
< 5
> cars
< ▶ (5) ['toyota', 'audi', 'bmw', 'xuv', 'maruti']
> cars.unshift("ferrari");
< 6
> cars
< ▶ (6) ['ferrari', 'toyota', 'audi', 'bmw', 'xuv', 'maruti']
> cars.shift();
< 'ferrari'
> cars
< ▶ (5) ['toyota', 'audi', 'bmw', 'xuv', 'maruti']
> let followers = ["a", "b", "c"];
< undefined
> let blocked = followers.shift();
< undefined
> followers
< ▶ (2) ['b', 'c']
> blocked
< 'a'
>
```

APNA COLLEGE

APNA COLLEGE

Practice Qs

Qs. For the given start state of an array, change it to final form using methods.

start :
```
['january', 'july', 'march', 'august']
```

final :
```
['july', 'june', 'march', 'august']
```

JS

Practice Qs

Qs. For the given start state of an array, change it to final form using methods.

start : `['january', 'july', 'march', 'august']`

"july" "june"

final : `['july', 'june', 'march', 'august']`

```
months.shift()
months.shift()
months.unshift("june");
  "       "    "july";
```

JS

APNA COLLEGE

Array Methods

```
> let primary = ["red", "yellow", "blue"];
```

indexOf : returns index of something

```
> primary.indexOf("yellow");
<· 1
> primary.indexOf("green");
<· -1
> primary.indexOf("Yellow");
<· -1
```

includes : search for a value

```
> primary.includes("red");
<· true
> primary.includes("green");
<· false
```

JS

Array **Methods**

```
> let primary = ["red", "yellow", "blue"];
<· undefined
> let secondary = ["orange", "green", "violet"];
<· undefined
```

"a" + "b"

concat : merge 2 arrays → concatenate

```
> primary.concat(secondary);
<· ▶ (6) ['red', 'yellow', 'blue', 'orange', 'green', 'violet']
```

reverse : reverse an array

```
> primary.reverse();
<· ▶ (3) ['blue', 'yellow', 'red']
```

JS

/Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console Sources Network Performance

top Filter Default levels No Issues

```
> let primary = ["red", "yellow", "blue"];
< undefined
> let secondary = ["orange", "green", "violet"];
< undefined
> primary.concat(secondary);
< ▶ (6) ['red', 'yellow', 'blue', 'orange', 'green', 'violet']
> primary
< ▶ (3) ['red', 'yellow', 'blue']
> secondary
< ▶ (3) ['orange', 'green', 'violet']
> let allColors = primary.concat(secondary);
< undefined
> allColors
< ▶ (6) ['red', 'yellow', 'blue', 'orange', 'green', 'violet']
>
```

APNA COLLEGE

Array **Methods**

```
> let primary = ["red", "yellow", "blue"];
<· undefined
> let secondary = ["orange", "green", "violet"];
<· undefined
```

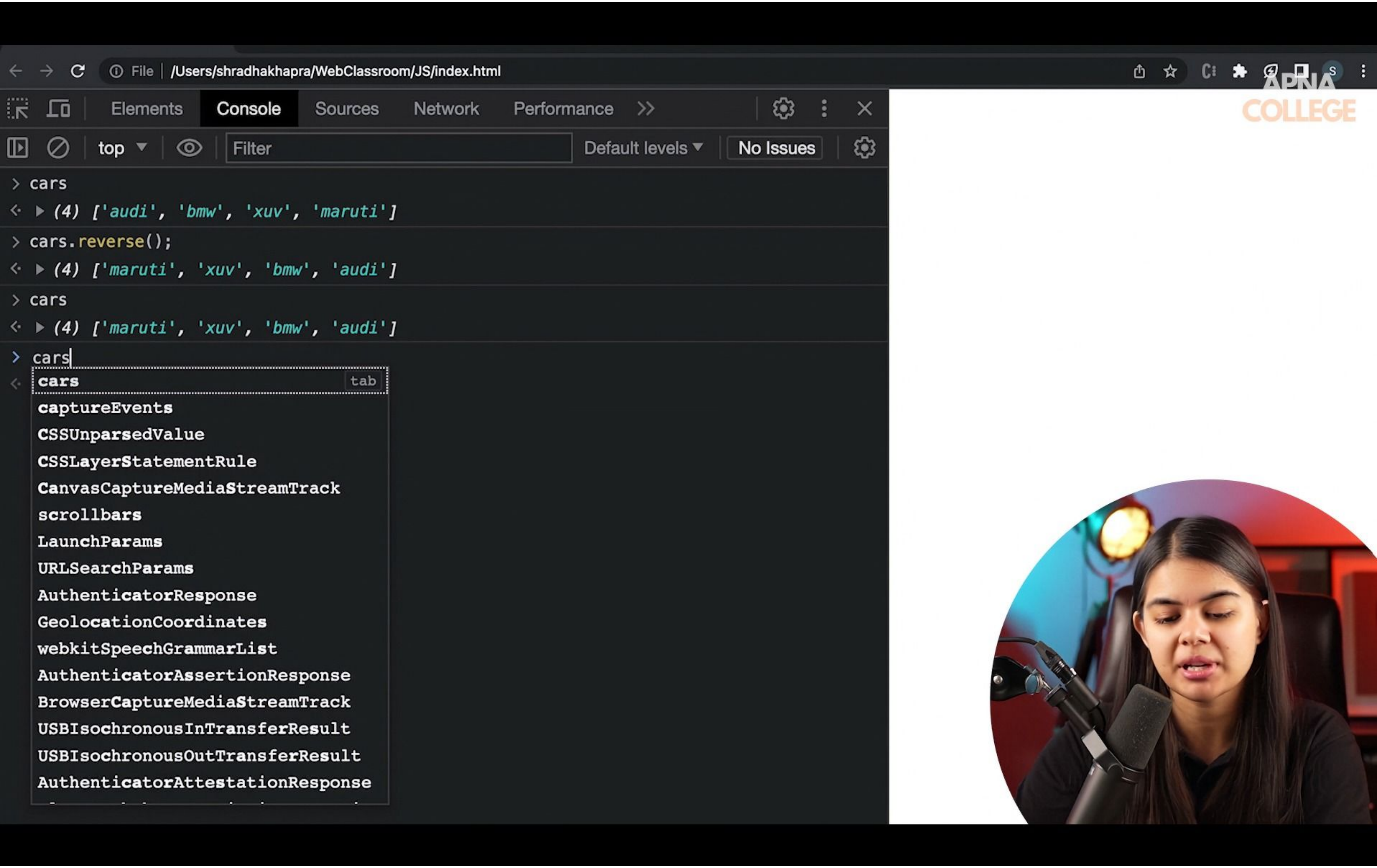
concat : merge 2 arrays → concatenate

```
> primary.concat(secondary);
<· ▶ (6) ['red', 'yellow', 'blue', 'orange', 'green', 'violet']
```

reverse : reverse an array

```
> primary.reverse();
<· ▶ (3) ['blue', 'yellow', 'red']
```

JS


File | /Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
APNA COLLEGE
> cars
< (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.reverse();
< (4) ['maruti', 'xuv', 'bmw', 'audi']
> cars
< (4) ['maruti', 'xuv', 'bmw', 'audi']
> cars
cars tab
captureEvents
CSSUnparsedValue
CSSLayerStatementRule
CanvasCaptureMediaStreamTrack
scrollbars
LaunchParams
URLSearchParams
AuthenticatorResponse
GeolocationCoordinates
webkitSpeechGrammarList
AuthenticatorAssertionResponse
BrowserCaptureMediaStreamTrack
USBIsochronousInTransferResult
USBIsochronousOutTransferResult
AuthenticatorAttestationResponse

slice (start , end)

APNA COLLEGE

Array **Methods**

slice : copies a portion of an array

```
> let colors = ["red", "yellow", "blue", "orange", "pink", "white"];
```

```
> colors.slice()
<· ▶ (6) ['red', 'yellow', 'blue', 'orange', 'pink', 'white']
> colors.slice(2);
<· ▶ (4) ['blue', 'orange', 'pink', 'white']
> colors.slice(2, 3);
<· ▶ ['blue']
> colors.slice(-2);
<· ▶ (2) ['pink', 'white']
```

JS

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top Filter Default levels No Issues

```
> cars
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.slice(-1)
< ▶ ['maruti']
> cars.slice(-2)
< ▶ (2) ['xuv', 'maruti']
> cars.slice(-3)
< ▶ (3) ['bmw', 'xuv', 'maruti']
> cars.slice(-4)
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.slice(-5)
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
>
```

APNA COLLEGE

Array Methods

```
> let colors = ["red", "yellow", "blue", "orange", "pink", "white"];
```

splice : removes / replaces / add elements in place

splice(start, deleteCount, item0...itemN)

OP

```
> colors.splice(4);
<· ▶ (2) ['pink', 'white']
> colors
<· ▶ (4) ['red', 'yellow', 'blue', 'orange']
> colors.splice(0, 1);
<· ▶ ['red']
> colors
<· ▶ (3) ['yellow', 'blue', 'orange']
> colors.splice(0, 1, "black", "grey");
<· ▶ ['yellow']
> colors
<· ▶ (4) ['black', 'grey', 'blue', 'orange']
```


JS

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Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> cars
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> cars.splice(3);
< ▶ ['maruti']
> cars
< ▶ (3) ['audi', 'bmw', 'xuv']
> cars.splice(0,1);
< ▶ ['audi']
> cars
< ▶ (2) ['bmw', 'xuv']
> cars.push("maruti");
  cars.push("ferrari");
< 4
> cars
< ▶ (4) ['bmw', 'xuv', 'maruti', 'ferrari']
> cars.splice(1, 2);
```

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Elements | Console | Sources | Network | Performance

```
<· ▶ (3) ['audi', 'bmw', 'xuv']
> cars.splice(0,1);
<· ▶ ['audi']
> cars
<· ▶ (2) ['bmw', 'xuv']
> cars.push("maruti");
  cars.push("ferrari");
<· 4
> cars
<· ▶ (4) ['bmw', 'xuv', 'maruti', 'ferrari']
> cars.splice(1, 2);
<· ▶ (2) ['xuv', 'maruti']
> cars
<· ▶ (2) ['bmw', 'ferrari']
> cars.splice(0, 0, "toyota", "xuv", "maruti");
<· ▶ []
> cars
<· ▼ (5) ['toyota', 'xuv', 'maruti', 'bmw', 'ferrari']
    0: "toyota"
    1: "xuv"
    2: "maruti"
    3: "bmw"
    4: "ferrari"
    length: 5
  ▶ [[Prototype]]: Array(0)
```

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top Filter Default levels No Issues

```
> cars
< ▶ (5) ['toyota', 'xuv', 'maruti', 'bmw', 'ferrari']
> cars.splice(1, 0, "mercedes");
< ▶ []
> cars
< ▶ (6) ['toyota', 'mercedes', 'xuv', 'maruti', 'bmw', 'ferrari']
> cars.splice(1, 1, "gwagon");
< ▶ ['mercedes']
> cars
< ▶ (6) ['toyota', 'gwagon', 'xuv', 'maruti', 'bmw', 'ferrari']
>
```

COLLEGE

APNA COLLEGE

Array Methods

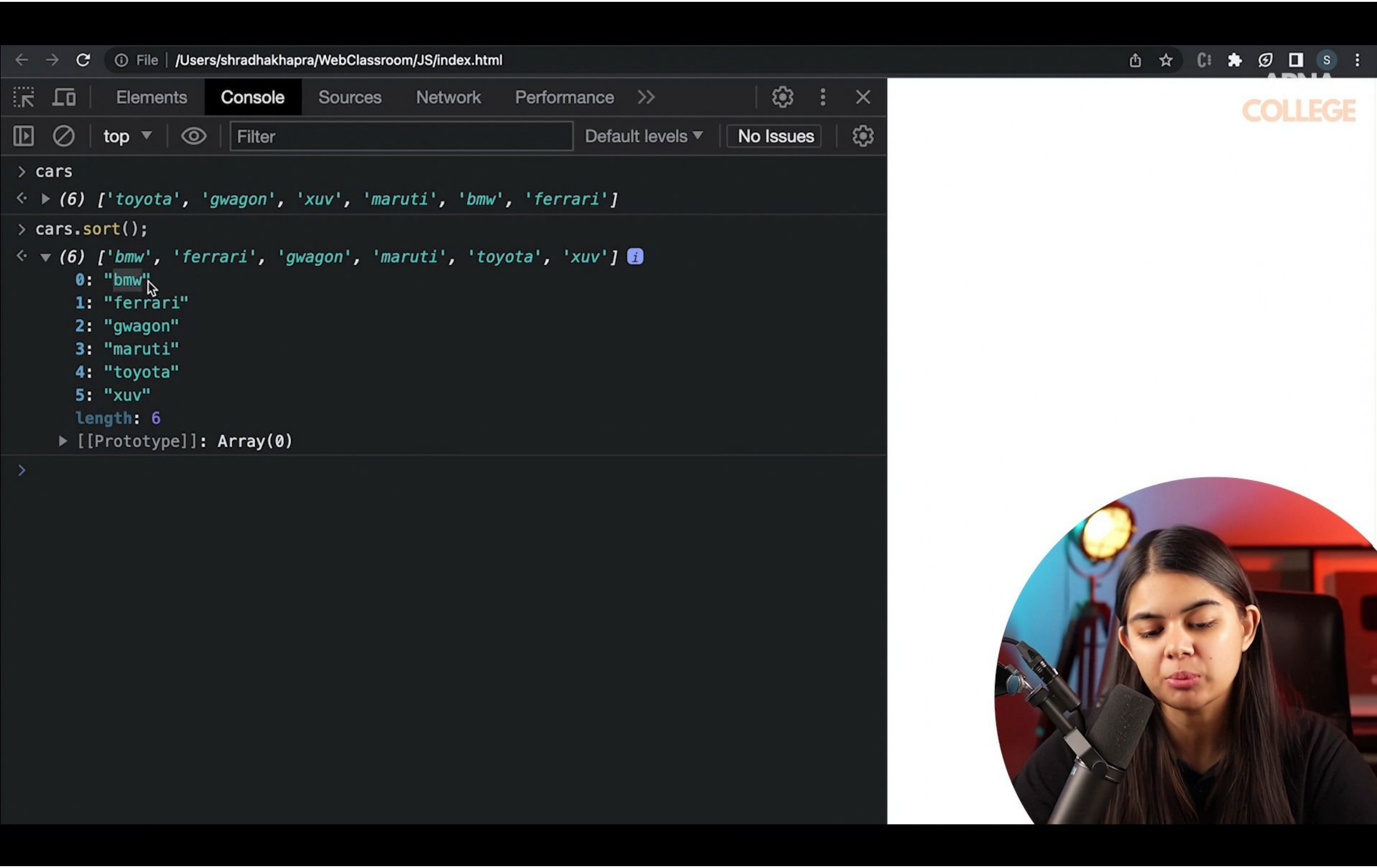
ascending
descending

sort : sorts an array

```
> let days = ["monday", "sunday", "wednesday", "tuesday"];
<· undefined
> days.sort();
<· ▶ (4) ['monday', 'sunday', 'tuesday', 'wednesday']
```

```
> let squares = [25, 16, 4, 49, 36, 9]
<· undefined
> squares.sort();
<· ▶ (6) [16, 25, 36, 4, 49, 9]
```

JS


/Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
COLLEGE
> cars
< (6) ['toyota', 'gwagon', 'xuv', 'maruti', 'bmw', 'ferrari']
> cars.sort();
< (6) ['bmw', 'ferrari', 'gwagon', 'maruti', 'toyota', 'xuv']
0: "bmw"
1: "ferrari"
2: "gwagon"
3: "maruti"
4: "toyota"
5: "xuv"
length: 6
[[Prototype]]: Array(0)

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Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> cars
< ▶ (6) ['toyota', 'gwagon', 'xuv', 'maruti', 'bmw', 'ferrari']
> cars.sort();
< ▶ (6) ['bmw', 'ferrari', 'gwagon', 'maruti', 'toyota', 'xuv']
> let chars = ['b', 'd', 'e', 'a'];
< undefined
> chars.sort();
< ▶ (4) ['a', 'b', 'd', 'e']
> let marks = [99, 89, 67, 42, 100];
< undefined
> marks.sort();
< ▶ (5) [100, 42, 67, 89, 99]
>
```

COLLEGE

03:14 / 03:16

shift
"
unshift
"] 4 steps

APNA COLLEGE

Practice Qs

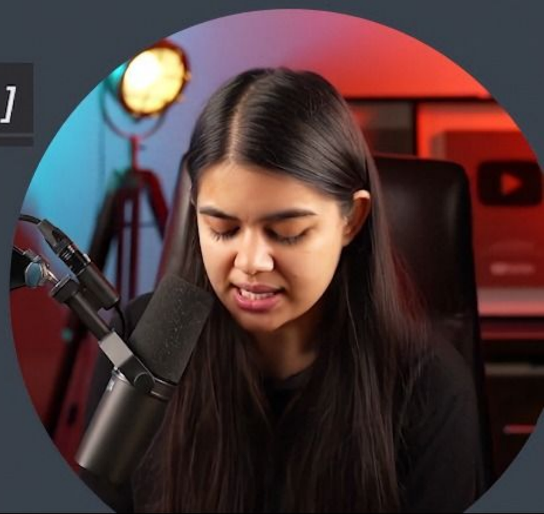
Qs. For the given start state of an array, change it to final form using splice.

start :
```
['january', 'july', 'march', 'august']
```

final :
```
['july', 'june', 'march', 'august']
```

Qs. Return the index of the "javascript" from the given array, if it was reversed.

```
['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
```



JS

00:32 / 04:24

APNA COLLEGE

shift
"
unshift
"] 4 steps

Practice Qs

Qs. For the given start state of an array, change it to final form using splice.

start : `['january', 'july', 'march', 'august']`

final : `['july', 'june', 'march', 'august']`

Qs. Return the index of the "javascript" from the given array, if it was reversed.

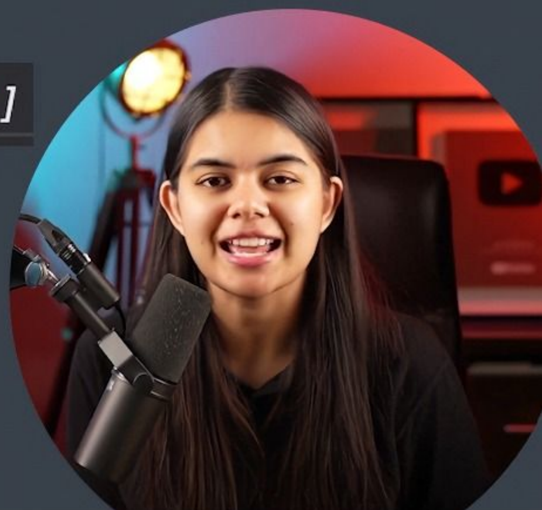
```
['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
```

0 1 2 3 4 5 6 7

7 6 5 4 3 2 1 0

lang.reverse()

JS



shift
"
unshift
"
4 steps

APNA COLLEGE

Practice Qs

Qs. For the given start state of an array, change it to final form using splice.

start : `['january', 'july', 'march', 'august']`

final : `['july', 'june', 'march', 'august']`

Qs. Return the index of the "javascript" from the given array, if it was reversed.

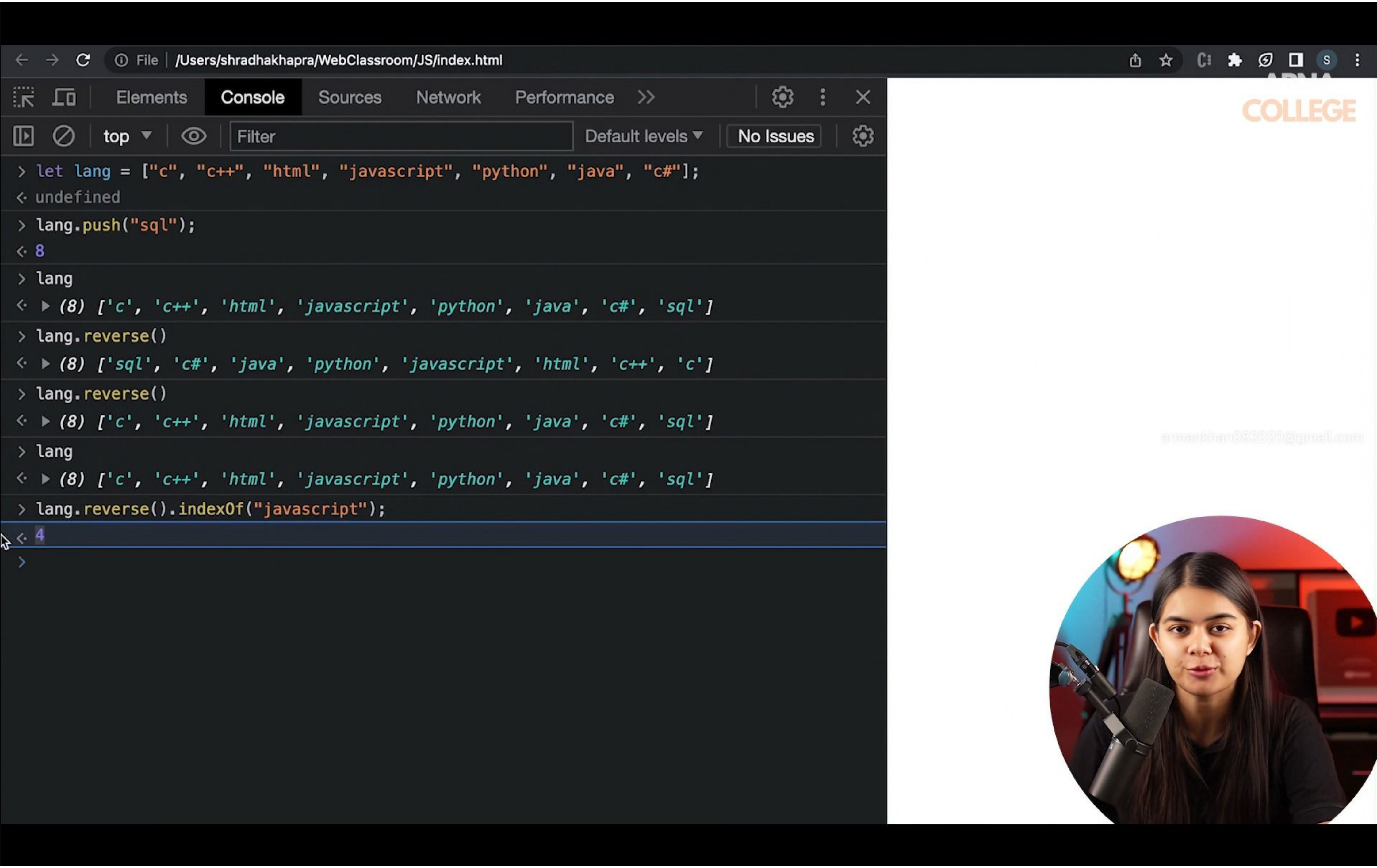
```
['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
```

0 1 2 3 4 5 6 7

7 6 5 4 3 2 1 0

lang.reverse().indexOf.

JS


File | /Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No Issues
> let lang = ["c", "c++", "html", "javascript", "python", "java", "c#"];
< undefined
> lang.push("sql");
< 8
> lang
< (8) ['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
> lang.reverse()
< (8) ['sql', 'c#', 'java', 'python', 'javascript', 'html', 'c++', 'c']
> lang.reverse()
< (8) ['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
> lang
< (8) ['c', 'c++', 'html', 'javascript', 'python', 'java', 'c#', 'sql']
> lang.reverse().indexOf("javascript");
< 4
COLLEGE

Array References

address in memory

```
> [1] === [1]
< false
> [1] == [1]
< false
```

JS

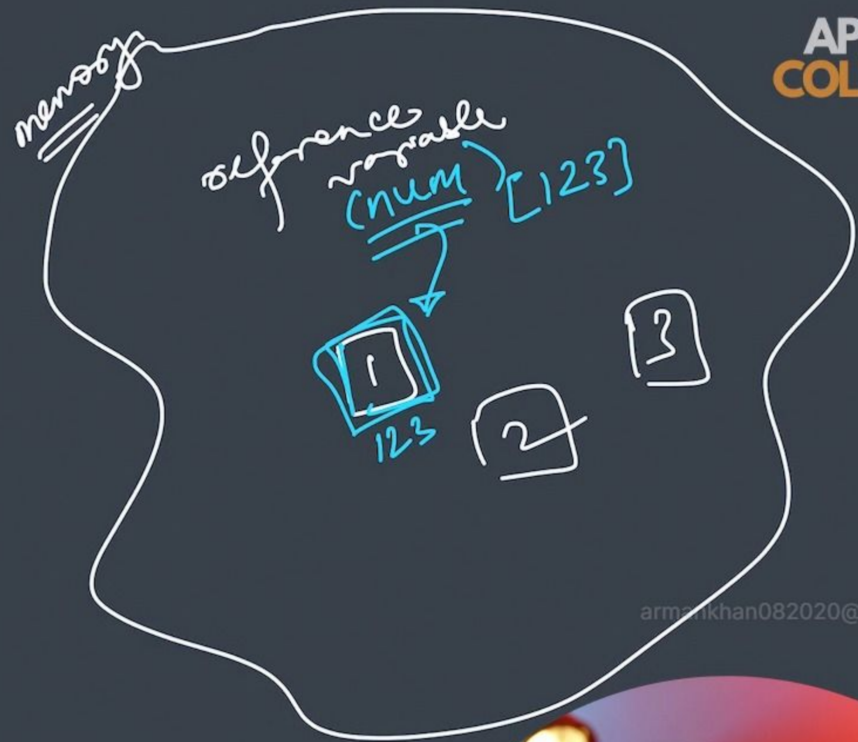
Array References

```
> [1] === [1]
<· false
> [1] == [1]
<· false
```

address in memory

let num = [1, 2, 3]

ref var



JS

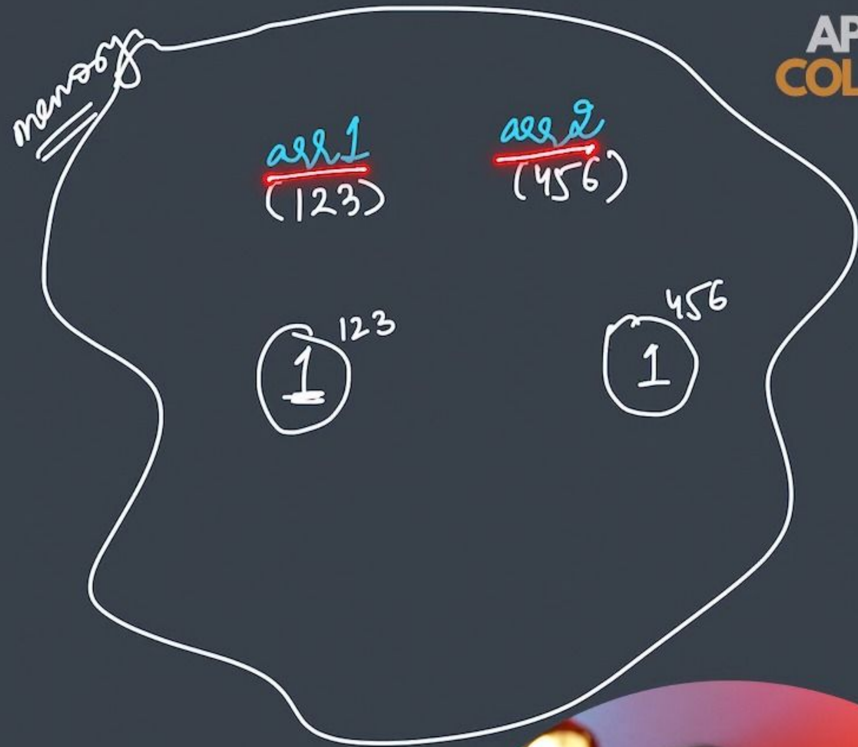
Array References

address in memory

```
> [1] === [1]
< false
> [1] == [1]
< false
```

let arr1 = [1] ✓

let arr2 = [1] ✓



JS

Array References

```
> let arr = ['a', 'b'];
<· undefined
> let arrCopy = arr;
<· undefined
> arrCopy
<· ▶ (2) ['a', 'b']
> arrCopy.push('c');
<· 3
> arr
<· ▶ (3) ['a', 'b', 'c']
```

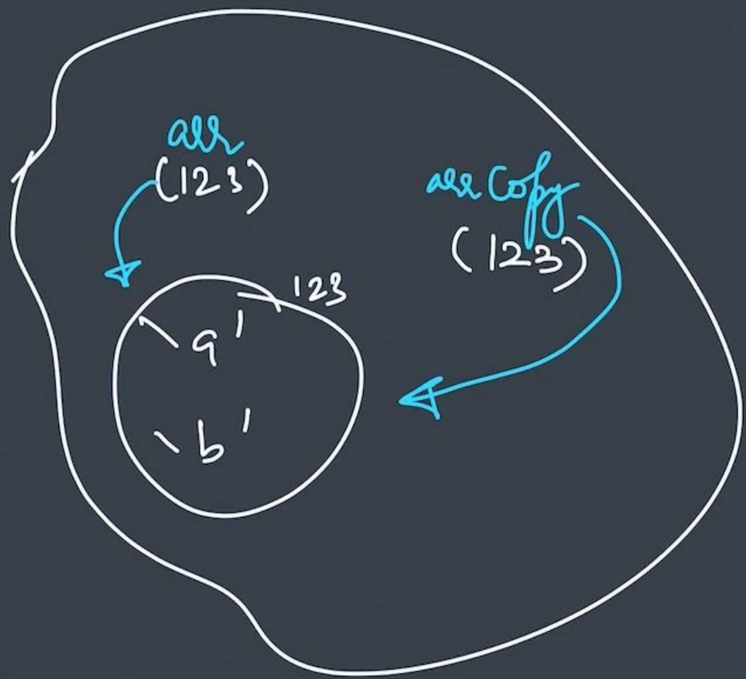
```
> arr == arrCopy
<· true
```

JS

Array References

```
> let arr = ['a', 'b'];
<· undefined
> let arrCopy = arr;
<· undefined
> arrCopy
<· ▶ (2) ['a', 'b']
> arrCopy.push('c');
<· 3
> arr
<· ▶ (3) ['a', 'b', 'c']
```

```
> arr == arrCopy
<· true
```



JS

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```
> let arr = ['a', 'b', 'c'];
<· undefined
> let arrCopy = arr;
<· undefined
> arr == arrCopy
<· true
> arr === arrCopy
<· true
>
```

COLLEGE

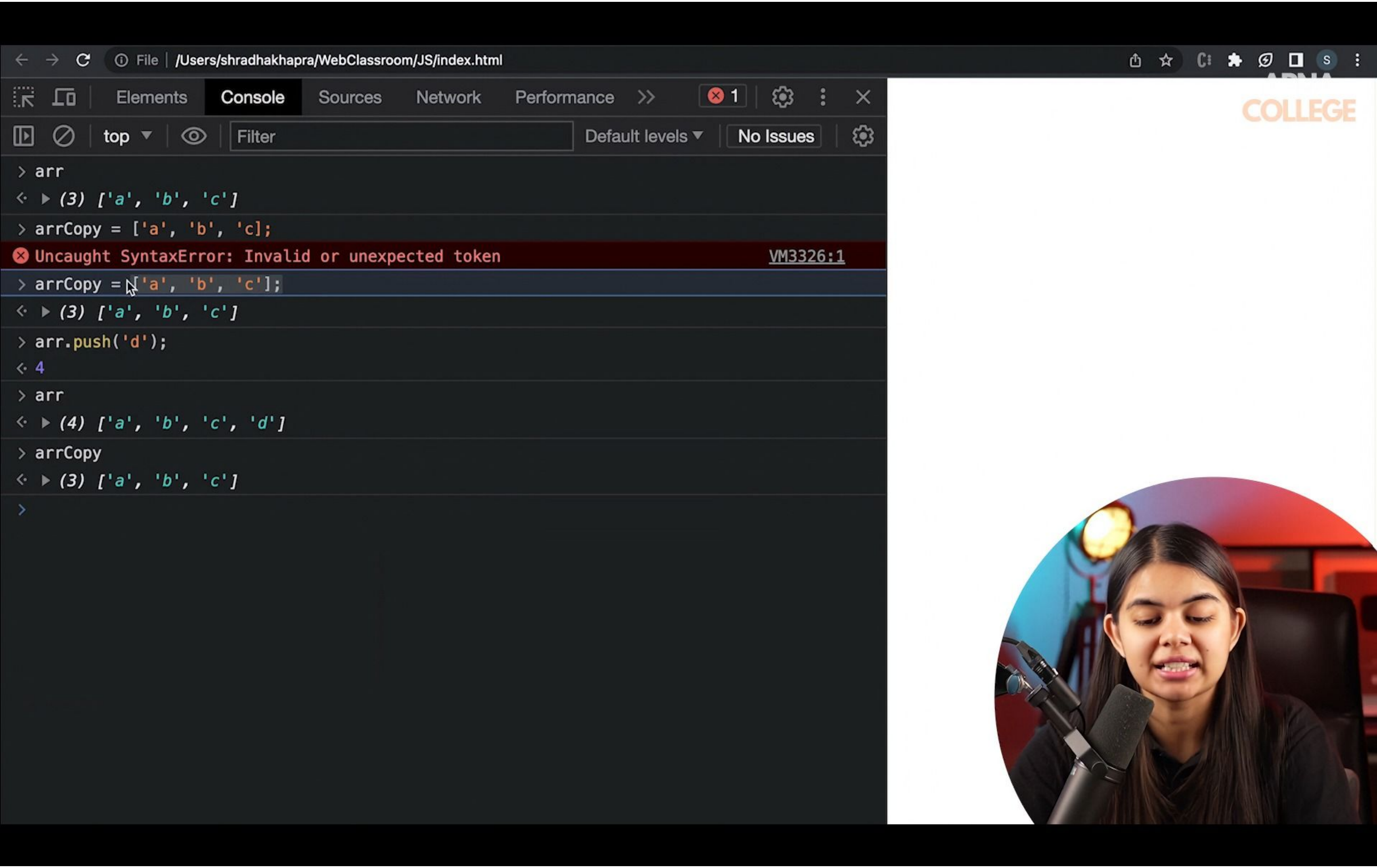
File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> let arr = ['a', 'b', 'c'];
< undefined
> let arrCopy = arr;
< undefined
> arr == arrCopy
< true
> arr === arrCopy
< true
> arr.push('d');
< 4
> arr
< ▶ (4) ['a', 'b', 'c', 'd']
> arrCopy
< ▶ (4) ['a', 'b', 'c', 'd']
> arrCopy.pop();
< 'd'
> arrCopy
< ▶ (3) ['a', 'b', 'c']
> arr
< ▶ (3) ['a', 'b', 'c']
>
```

COLLEGE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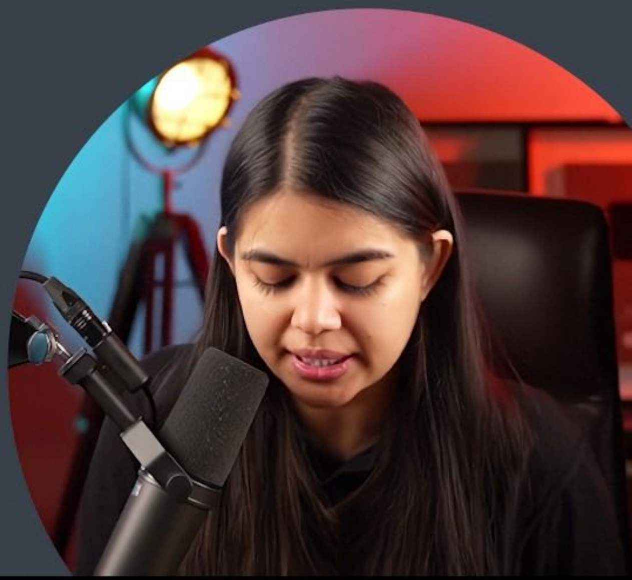
/Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
1
top
Filter
Default levels
No Issues
COLLEGE
> arr
<· ▶ (3) ['a', 'b', 'c']
> arrCopy = ['a', 'b', 'c];
Uncaught SyntaxError: Invalid or unexpected token
VM3326:1
> arrCopy = ['a', 'b', 'c'];
<· ▶ (3) ['a', 'b', 'c']
> arr.push('d');
<· 4
> arr
<· ▶ (4) ['a', 'b', 'c', 'd']
> arrCopy
<· ▶ (3) ['a', 'b', 'c']

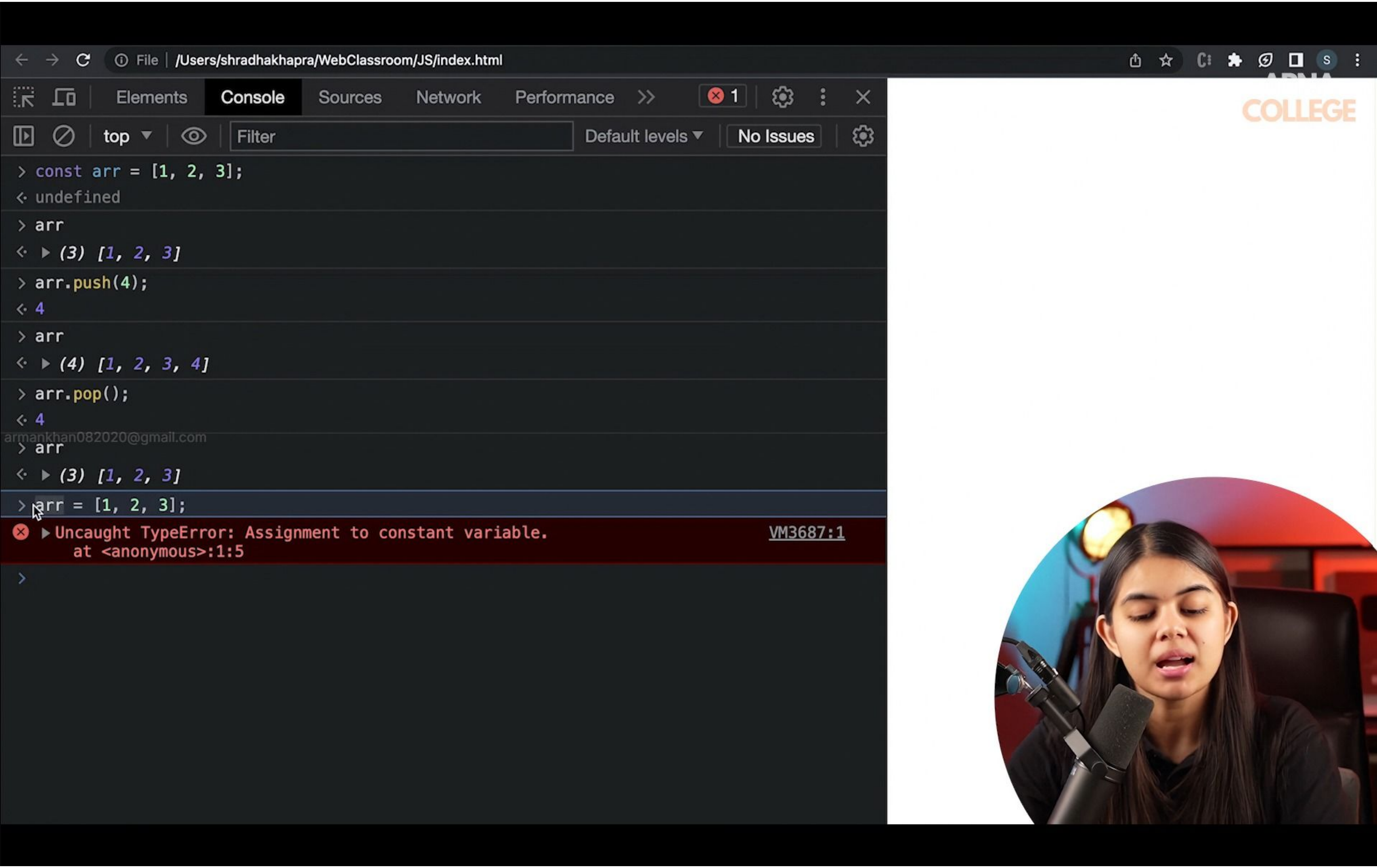
Constant Arrays

```
const arr = [1, 2, 3, 4];
```

```
let a = 5;

const a = 5
```




File | /Users/shradhakhapra/WebClassroom/JS/index.html
Elements
Console
Sources
Network
Performance
1
top
Filter
Default levels
No Issues
COLLEGE
> const arr = [1, 2, 3];
< undefined
> arr
< (3) [1, 2, 3]
> arr.push(4);
< 4
> arr
< (4) [1, 2, 3, 4]
> arr.pop();
< 4
armankhan082020@gmail.com
> arr
< (3) [1, 2, 3]
> arr = [1, 2, 3];
Uncaught TypeError: Assignment to constant variable.
at <anonymous>:1:5
VM3687:1

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Constant Arrays

```
const arr = [1, 2, 3, 4];
```

arr constant

address

let a = 5;

const a = 5

[1, 2, 3, ~~4~~]

5, 6

JS

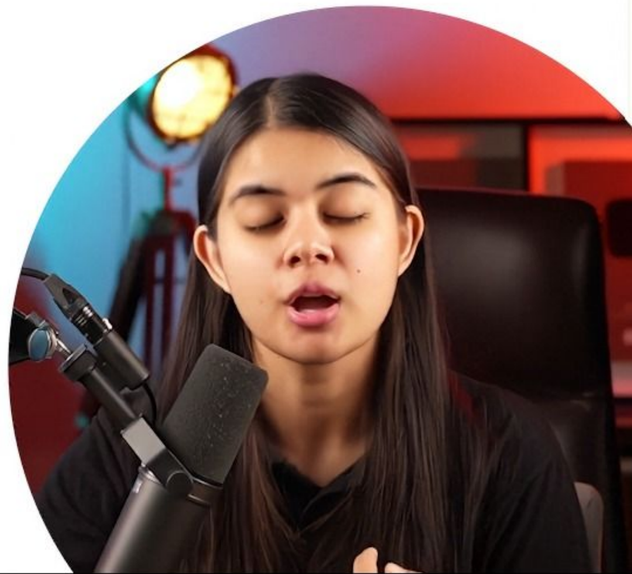
File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements | Console | Sources | Network | Performance | 2

top | Filter | Default levels | No Issues

```
> const arr = [1, 2, 3];
< undefined
> arr
< ▶ (3) [1, 2, 3]
> arr.push(4);
< 4
> arr
< ▶ (4) [1, 2, 3, 4]
> arr.pop();
< 4
> arr
< ▶ (3) [1, 2, 3]
> arr = [1, 2, 3];
✖ ▶Uncaught TypeError: Assignment to constant variable.      VM3687:1
     at <anonymous>:1:5
> cars
< ▶ (4) ['audi', 'bmw', 'xuv', 'maruti']
> arr = cars;
✖ ▶Uncaught TypeError: Assignment to constant variable.      VM3728:1
     at <anonymous>:1:5
>
```

COLLEGE



Nested Arrays

array of arrays

```
> let nums = [ [2, 4], [3, 6], [4, 8] ];
< undefined
> nums
< ▼ (3) [Array(2), Array(2), Array(2)] i
    ▶ 0: (2) [2, 4]
    ▶ 1: (2) [3, 6]
    ▶ 2: (2) [4, 8]
      length: 3
    ▶ [[Prototype]]: Array(0)
```

JS

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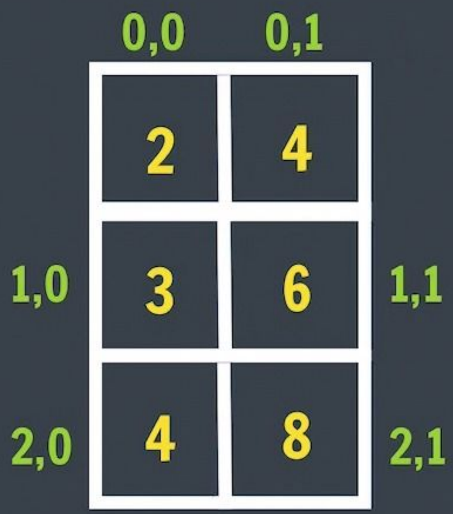
Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> let nums = [ [1, 2], [3, 4], [4, 5] ];
< undefined
> let nums = [ [2, 4], [3, 6], [4, 8] ];
< undefined
> nums
< ▼ (3) [Array(2), Array(2), Array(2)]
    ▶ 0: (2) [2, 4]
    ▶ 1: (2) [3, 6]
    ▶ 2: (2) [4, 8]
      length: 3
    ▶ [[Prototype]]: Array(0)
>
```

Nested Arrays

```
> let nums = [ [2, 4], [3, 6], [4, 8] ];
```




```
nums[0][0]       //2
```

JS

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Elements | Console | Sources | Network | Performance

top | Filter | Default levels | No Issues

```
> let nums = [ [1, 2], [3, 4], [4, 5] ];
<· undefined
> let nums = [ [2, 4], [3, 6], [4, 8] ];
<· undefined
> nums
<· ▶ (3) [Array(2), Array(2), Array(2)]
> nums.length
<· 3
> nums[0]
<· ▶ (2) [2, 4]
> nums[0].length
<· 2
> nums[0][0]
<· 2
> nums[1][1]
<· 6
> nums[1][2]
<· undefined
> nums[2][0]
<· 4
>
```

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1D

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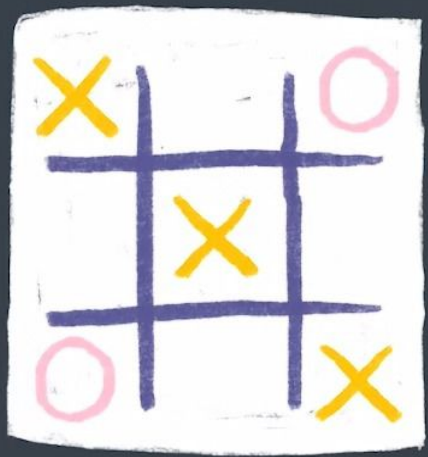
Practice Qs

Qs. Create a nested array to show the following tic-tac-toe game state.

2D array



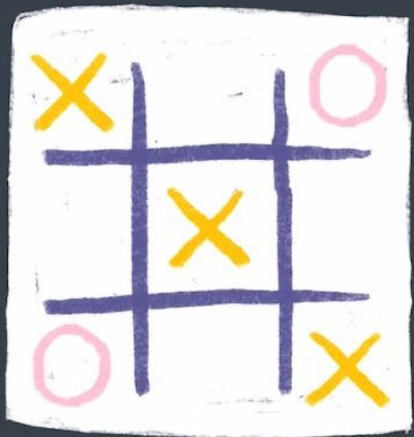
S

1D

APNA COLLEGE

Practice Qs

Qs. Create a nested array to show the following tic-tac-toe game state.



2D array

'X'	null	'O'
null	'X'	null
'O'	null	'X'

[['X', null, 'O'], [null, 'X', null], ['O', null, 'X']]

1 2

JS

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Elements Console Sources Network Performance

top Filter Default levels No Issues

```
> let game = [ ['X', null, 'O'], [null, 'X', null], ['O', null, 'X'] ];
<· undefined
> game
<· ▼ (3) [Array(3), Array(3), Array(3)]
    ▶ 0: (3) ['X', null, 'O']
    ▶ 1: (3) [null, 'X', null]
    ▶ 2: (3) ['O', null, 'X']
      length: 3
    ▶ [[Prototype]]: Array(0)
> game[0]
<· ▶ (3) ['X', null, 'O']
> game[0][1]
<· null
> game[0][1] = 'O';
<· 'O'
> game
<· ▼ (3) [Array(3), Array(3), Array(3)]
    ▶ 0: (3) ['X', 'O', 'O']
    ▶ 1: (3) [null, 'X', nu Array(3)
    ▶ 2: (3) ['O', null, 'X']
      length: 3
    ▶ [[Prototype]]: Array(0)
>
```

COLLEGE

APNA COLLEGE

JS (Part 3)

Practice Questions

Qs1. Write a JavaScript program to get the first n elements of an array. [n can be any positive number].
For example: for array [7, 9, 0, -2] and n=3
Print, [7, 9, 0]

Qs2. Write a JavaScript program to get the last n elements of an array. [n can be any positive number].
For example: for array [7, 9, 0, -2] and n=3
Print, [9, 0, -2]

Qs3. Write a JavaScript program to check whether a string is blank or not.

Qs4. Write a JavaScript program to test whether the character at the given (character) index is lower case.

Qs5. Write a JavaScript program to strip leading and trailing spaces from a string.

Qs6. Write a JavaScript program to check if an element exists in an array or not.